NMIMS · MBA TECH · FINANCIAL TECHNOLOGY (703TF0E006)

FinTech Midterm Study Guide

Units 1 to 7: every concept, framework, case study and number from the lectures and Michael King's *Fintech Explained*, laid out in one flow so it is easy to revise and recall.

UNIT 1
Introduction to FinTech

UNIT 2
Market Evolution & Current Trends

UNIT 3
AI & Machine Learning in Finance

UNIT 4
Blockchain & Cryptocurrencies

UNIT 5
Payment Systems & Digital Wallets

UNIT 6
P2P Lending & Crowdfunding

UNIT 7
Robo-Advisors, Open Banking & APIs

FINAL
One-page cheat sheet & checklist

Source: Michael R. King, *Fintech Explained: How Technology Is Transforming Financial Services* (University of Toronto Press, 2023), plus the Unit 1–7 lecture decks.

How to use this guide

Each unit follows the same pattern, so you always know where to look:

1 · One-line idea
The whole unit in one sentence. If you remember nothing else, remember this.

2 · Concepts + diagrams
Frameworks explained in plain words, each with a picture.

3 · Cases + numbers
The companies and figures the lecturer used, ready to drop into answers.

4 · Recall + practice
Mnemonic, exam traps, model answers and a 60-second self-test.

Boxes are colour-coded: KEY NUMBERS MNEMONIC EXAM TRAP ADDED FROM SLIDES. The **Added from slides** boxes hold material that was in the lecture decks but missing from your first draft. TIER 1 · HIGH WEIGHT and TIER 2 · CORE tags show how heavily each unit is likely to be tested.

The story of the course

The seven units tell one story. Keep it in your head and every topic has a place to sit:

Unit 1 What is FinTech? Lens · paradigms → **Unit 2** Why it disrupts economics & strategy → **Unit 3** AI & ML data → decisions → **Unit 4** Blockchain trust w/o middleman → **Unit 5** Payments moving money → **Unit 6** P2P & crowd raising money → **Unit 7** Robo + Open Banking growing & sharing

Units 1–2 give you the **lenses**; Units 3–4 are the **technologies**; Units 5–7 are the **business lines** you analyse with those lenses.

FIVE IDEAS THAT APPEAR IN EVERY UNIT (USE THEM IN ANY ANSWER)

1. **The FinTech Explained Lens**: Customer → Pain point → Value proposition → Monetisation (Unit 1).
2. **Five dimensions of intermediation**: information asymmetry, transaction costs, liquidity creation, risk sharing, trust (Unit 2).
3. **Agency vs principal revenue**: fees without balance-sheet risk vs interest spread with risk (Units 2, 6).
4. **Multi-sided platforms**: subsidy side gets it free, money side pays; cross-side network effects (Units 2, 5, 7).
5. **Unbundle → rebundle → partner**: start-ups carve off one product, rebundle into super-apps, then often partner with or become banks (Units 2, 6, 7).

HOW TO WRITE A 10-MARK ANSWER

Define the concept → name the framework → apply it step by step to the case → use one or two numbers → give a trade-off (strengths and risks) → finish with a one-line recommendation. Every model answer in this guide follows that order.

01 Introduction to FinTech

Course outcomes: CO1 (fundamentals) · CO5 (apply to ventures)

TIER 2 · CORE

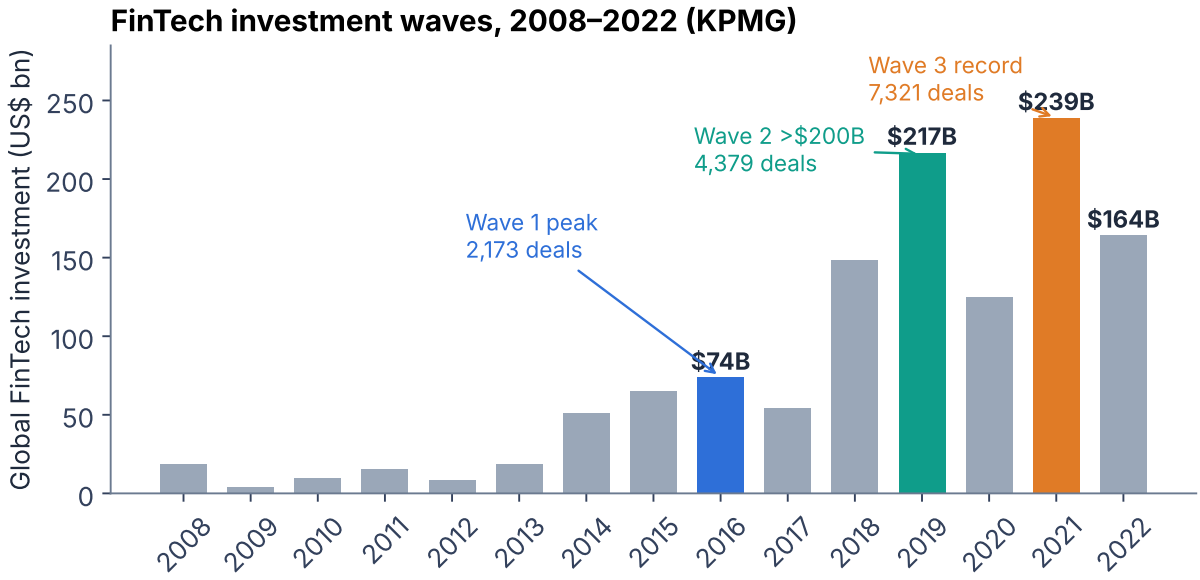
In one line: FinTech is **not** just an IT upgrade for banks. It is a **customer-centric** shift that delivers financial services **cheaper, easier, faster and more conveniently** through software, mobile, cloud and data.

1.1 Definition and value proposition

FinTech = the digital delivery of financial products and services using software, cloud infrastructure, mobile networks and the internet. It moves finance away from **product-centric branch banking** towards **customer-centric digital ecosystems** that remove friction and transaction costs.

The value test is always the same four words: **cheaper · easier · faster · more convenient**.

1.2 Investment waves (why FinTech exploded)



King Fig. 1.1 (KPMG *Pulse of Fintech*): over $1.2 trillion across ~38,000 deals, 2008–2022.

Wave	Period	What happened
Wave 1	post-GFC → 2016	Rebounded from the 2012 low to a peak of **$73.7B across 2,173 deals** in 2016.
Wave 2	2018–2019	Passed **$200B in 2019 (4,379 deals)**, then slowed in the 2020 COVID year.
Wave 3	2021 peak, 2022 reset	Record **~$240B, 7,321 deals** in 2021, then fell to **$164B** in 2022 as interest rates and the cost of capital rose.

Creative destruction (Schumpeter): most early start-up equity is lost to the high failure rate, while later funding rounds go to proven scale-ups. Media attention followed the money: mentions of "fintech" rose from about 191 (2000) to **225,167 (2018)**.

1.3 FinTech 1.0 → 2.0 → 3.0

1858 – LATE 1960s

FinTech 1.0 · Analog

Continuous electrical signals. 1858 transatlantic cable; 1867 stock ticker sends market prices across borders.

LATE 1960s – MID 2000s

FinTech 2.0 · Digital

Mainframes, PCs, networks. ATM (1967), Nasdaq (1971), mobile phone (1973), WWW (1989), online banking, 2G/3G. **Upgraded incumbents and wholesale markets.**

2000s – PRESENT

FinTech 3.0 · Start-ups

Democratising finance for retail consumers and small businesses. LendingClub & Zopa (2006), Kickstarter (2008), Bitcoin (2009).

The four catalysts of FinTech 3.0

☁ Cloud (2006)
AWS made computing and storage cheap and on-demand, with no big upfront capital needed.

📱 Smartphones (2007–08)
iPhone (2007) and Android (2008) put a computer in every pocket, giving 24/7 access.

🏦 2008–09 GFC
Trust in banks collapsed; experienced bankers and techies left to build start-ups.

👥 Millennials
Digital natives trust Apple and Google more than banks and expect great UX.

MNEMONIC · CATALYSTS

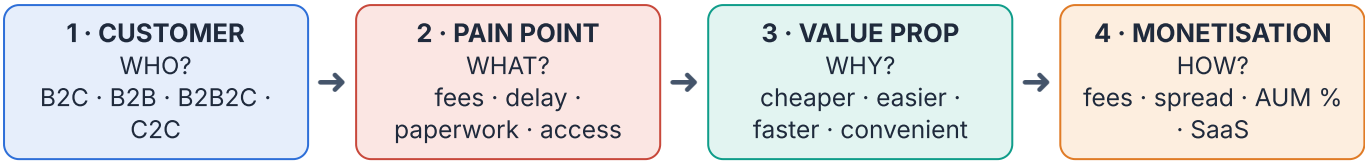
"C-S-G-M": Cloud, Smartphone, GFC, Millennials. Think "Cloud Smartphones Gave Millennials banking".

1.4 Two competing paradigms

Dimension	Traditional (Evolutionary)	Transformational (Revolutionary)
Core view	Natural, gradual evolution of banking	Disruptive revolution overturning an oligopolistic system
Focus	Product-centric (siloed: loans, deposits, cards)	Customer-centric (journeys, user experience)
Role of tech	Automation to cut back-office costs and protect rents (e.g. net interest margin)	Enabler that removes user pain points and friction
Business model	One-sided, centralised, legacy channels	Multi-sided platforms, DeFi, ecosystems
End goal	Maximise incumbent shareholder and insider profit	Cheaper, faster, easier, accessible finance

The two paradigms **coexist**: incumbents run the evolutionary playbook while start-ups run the revolutionary one, and many end up partnering.

1.5 The FinTech Explained Lens (the most-used framework)



King Fig. 1.3: the four questions you ask about any FinTech. It works as the elevator pitch for any venture.

1. **Customer (Who?)**: the target or underserved segment (unbanked, overcharged retail, freelancers, SMEs). State the segment type: B2C, B2B, B2B2C or C2C.
2. **Pain point (What?)**: the friction: opaque fees, high cost, slow settlement, paperwork, access hurdles.
3. **Value proposition (Why?)**: measurable advantage: **cheaper, easier, faster, more convenient**.
4. **Monetisation (How?)**: how value is captured: transaction fees, FX spread, interchange, origination fees, % of AUM, SaaS subscriptions, platform commissions.

The Lens in action (lecture examples)

Wise (cross-border payments)

- **Who** Expats, foreign students, digital nomads, cross-border SMEs
- **What** Hidden bank FX mark-ups; SWIFT takes 3–5 days
- **Why** Real mid-market rate, low upfront fee, fast
- **How** Small % fee per transfer, on volume

SoFi (lending & banking)

- **Who** HENRYs ("High Earners, Not Rich Yet"), graduates
- **What** High rates and rigid terms on student debt
- **Why** Low-rate refinancing, digital underwriting, community
- **How** Origination fees, interest spread, cross-selling

Wealthsimple (digital wealth)

- **Who** Young professionals below advisor minimums ($100k–250k+)
- **What** 1.5–2% fees and high minimums
- **Why** Automated low-cost ETF portfolios, simple app
- **How** 35–50 bps of AUM per year

MNEMONIC · THE LENS

C ustomer – who? | **P** ain point – what? | **V** alue – why? | **M** onetisation – how?

"**C**an **P**eople **V**alue **M**oney?"

1.6 Who deals with whom, what they sell, who they are

Operating segments

Segment	Meaning	Examples
B2C	Sell straight to retail consumers	Neobanks, direct robo-advisors, payment apps
B2B	Sell software / infrastructure to banks and enterprises, often white-label or embedded	Plaid, Finastra, Zafin
B2B2C	Partner with institutions to reach consumers, keeping visibility of the customer	Credit Karma, Borrowell (comparison marketplaces)
C2C / P2P	Individuals transact directly, with no balance-sheet middleman	Crypto exchanges, P2P lending portals

ADDED FROM SLIDES · THE 8 CORE LINES OF FINTECH BUSINESS

1 Cryptocurrencies: digital tokens on cryptographic ledgers (Bitcoin)

2 DeFi: smart contracts and dApps that lend, borrow and trade with no intermediary

3 Online lending: automated consumer and SME credit (debt crowdfunding)

4 Crowdfunding: equity, rewards or donations from the crowd

5 Digital banking: neobanks / challenger banks

6 Digital wealth: robo-advisors, ETF investing

7 Payments: checkout, acquiring, P2P, remittances

8 InsurTech: digital policies, sensor underwriting, AI claims

Four types of FinTech company

Capability	Start-up	Mature / listed	Digital bank / incumbent	BigTech platform
Innovation culture	✔	✔	✘	✔
Technical expertise	✔	✔	✔	✔
Financial expertise	✘	✔	✔	✘
Regulatory / risk-management expertise	✘	✘	✔	✘
Economies of scale	✘	✔	✘	✔
Access to customers	✘	✔	✘	✔
Access to funding	✘	✘	✔	✔
Access to talent	✔	✘	✘	✔
Examples	~90% fail; high CAC	PayPal, Stripe, Adyen	Nubank, Chime, Marcus (Goldman)	Apple, Google, Amazon, Ant, Tencent

Read the table as: **BigTech has the most ticks except regulation and finance**, which is why it partners with banks. **Start-ups have only culture, tech and talent**, which is why they struggle with customer acquisition and funding.

The FinTech ecosystem

A multi-stakeholder network: **FinTech ventures** (creative destruction) · **innovation labs & hubs** (e.g. MaRS, Toronto) · **incubators & accelerators** (Y Combinator, Techstars, Level39) · **capital providers** (FFF: friends, family & fools, angels, VCs, PE, corporate VC) · **regulators** (safety plus **regulatory sandboxes**) · **infrastructure** (universities, cloud, open-banking APIs).

1.7 FinTech for social good and financial inclusion

KEY NUMBERS

~1.7 billion adults are unbanked (World Bank), and **6%** even in rich economies like the US · **over two-thirds** of them own a mobile phone · in the last decade **1.2 billion** adults gained an account, cutting the unbanked population by **35%**.

Group 1: profitable underserved provision
Use cheap mobile rails to bank the unbanked sustainably, e.g. **M-PESA** (Kenya, SMS mobile money).

Group 2: incumbent disintermediation
Ideologically driven to strip oligopoly rents, e.g. **Bitcoin** (no central bank) and **Wise** (radical FX transparency).

1.8 Foundational technologies (preview of later units)

Technology	One-line meaning	Unit
APIs	Software messengers that let apps share data securely. **Closed** = inside a bank; **open** = third parties under PSD2 / UK Open Banking.	7
AI / ML	Supervised (labelled data, e.g. default prediction), unsupervised (clusters), reinforcement (reward loops).	3
Big data	"Data is the new oil" (Clive Humby, 2006): raw data must be refined into decisions.	3
Blockchain / DLT	Shared, cryptographically secured ledger; solves double-spend without a clearing house.	4
Cloud & "as-a-service"	AWS / Google Cloud / Azure → **SaaS, PaaS, IaaS, BaaS** subscription models.	7
Cryptography / SHA-256	One-way hash → fixed 256-bit output that can't be reversed.	4
Ethereum & smart contracts	Programmable blockchain; "if A happens, execute B" → dApps, DAOs.	4
MSPs and P2P networks	Platforms linking user groups (indirect network effects); networks with no central server.	2, 6

1.9 Exam practice

5 marks **Explain the difference between the Traditional (Evolutionary) and Transformational (Revolutionary) paradigms, mapping each to its business-model architecture.** CO1

The **Traditional (Evolutionary) paradigm** sees FinTech as a gradual upgrade to banking. Banks stay **product-centric** and use digital channels (mobile apps, online portals) mainly to **cut back-office costs** and protect their net interest margin. The architecture is **one-sided and centralised**: one institution designs, underwrites, holds and sells its own products.

The **Transformational (Revolutionary) paradigm** sees FinTech as a disruption of an oligopolistic, rent-extracting system. It is **customer-centric**: it **unbundles** banking products to fix specific pain points and **disintermediates** incumbents. The architecture is a **multi-sided platform** that connects user groups directly (borrowers ↔ lenders, buyers ↔ merchants), pushes marginal transaction costs toward zero and creates **cross-side network effects**.

Conclusion: the two coexist. Incumbents follow the first, start-ups the second, and partnerships blend them.

10 marks **You are a VC analyst evaluating a start-up that automates cross-border supply-chain payments for SMEs. Apply the FinTech Explained Lens to structure your pitch evaluation.** CO1, CO5

1 · Customer (Who?)

SMEs that import and export. Tier-1 banks under-serve them, saving good FX pricing and relationship managers for large corporates. Segment: **B2B**.

2 · Pain point (What?)

- **Cost:** correspondent banks each deduct fees and add hidden FX spreads of **2–4%** above spot.
- **Delay:** SWIFT wires take **3–5 business days**, locking up working capital.
- **Admin:** manual invoice matching and cross-border tax paperwork.

3 · Value proposition (Why?)

- **Cheaper:** near mid-market FX via local accounts cuts FX cost by 60–80%.
- **Faster:** local payment rails instead of multi-hop SWIFT give same-day settlement.
- **Easier / more convenient:** APIs plug into QuickBooks / Xero and match invoices automatically.

4 · Monetisation (How?)

Dual revenue: a **volume fee** of 30–50 bps on payments plus a **tiered SaaS subscription** (multi-currency accounts, tax reporting).

Cost on a $100,000 FX transfer	Legacy bank	FinTech rail
FX spread mark-up	2.50% = $2,500	0.35% = $350
SWIFT / outward wire fee	$50	$0
Intermediary bank deductions	$75	$0
Platform SaaS fee (allocated)	$0	$25
Total cost	**$2,625**	**$375 → saves $2,250 (85.7%)**

5 · Trade-offs and recommendation

Strengths: sticky integration into accounting software; scalable digital distribution. **Risks:** depends on partner banks for local clearing; price competition if banks upgrade their portals. **Verdict:** proceed to due diligence, subject to verifying bank partnerships and compliance.

60-SECOND RECALL · UNIT 1

- ☐ Define FinTech and state the four-word value test.
- ☐ Name the three investment waves with their peak year and size.
- ☐ FinTech 1.0 / 2.0 / 3.0: one milestone each. The four catalysts (C-S-G-M).
- ☐ Draw the Lens and apply it to Wise in 30 seconds.
- ☐ B2C / B2B / B2B2C / C2C with one example each.
- ☐ Which company type lacks regulatory expertise but has customers and funding? (BigTech)

02 Financial Services Market Evolution & Current Trends

TIER 1 · HIGH WEIGHT

Course outcomes: CO1 (intermediation theory) · CO2 (role of technology)

In one line: Banks exist to fix **five frictions**. FinTech fixes the same five more cheaply with data and code. It enters at the **low end** (Christensen), often as a **multi-sided platform** earning **agency fees** instead of interest spreads.

2.1 What financial intermediaries do

Banks & credit unions
Take deposits, make loans (commercial, retail, mortgage banks).

Brokers & dealers
Help buy and sell stocks, bonds and other assets.

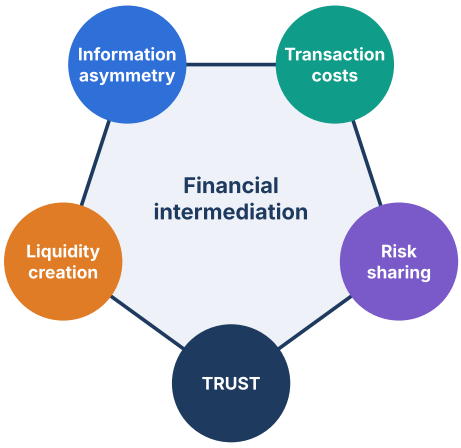
Investment banks
Help firms issue bonds or equity, or do M&A.

Asset managers
Pool savings: pension, mutual and hedge funds, PE, VC.

Insurers
Collect premiums and pay out when bad events happen.

Exchanges, clearing houses, depositories
Run the plumbing of financial markets.

2.2 The five dimensions of financial intermediation



King Fig. 2.1: trust sits at the base.

MNEMONIC · "I-T-L-R-T"

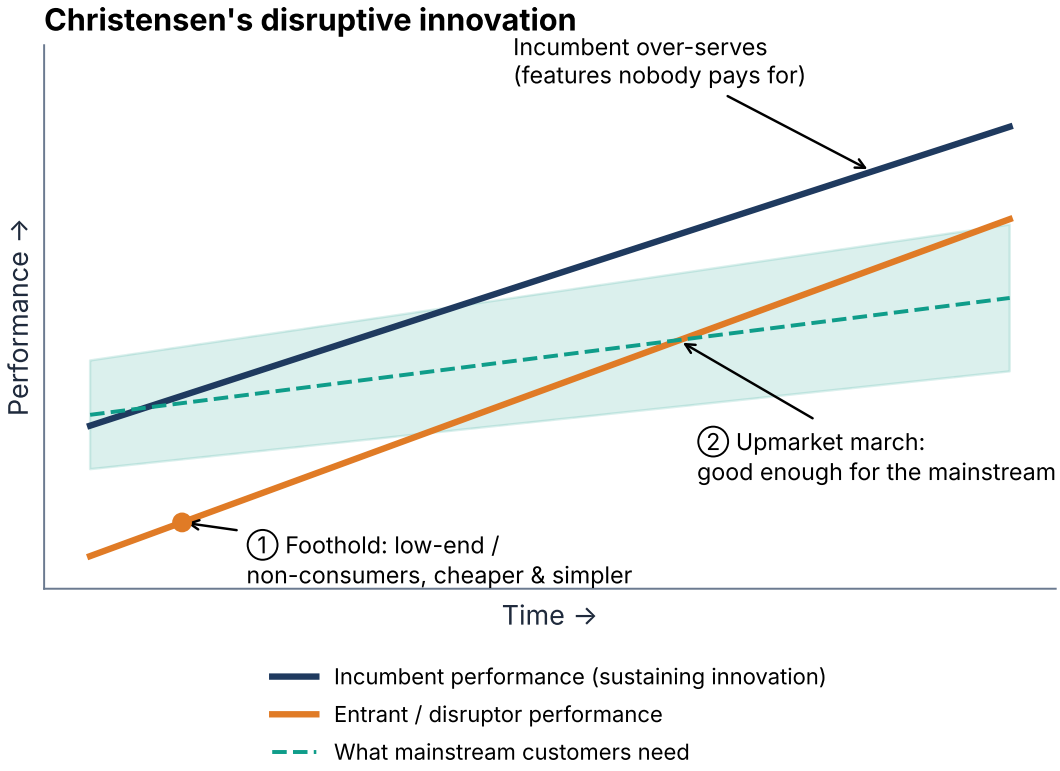
I nformation asymmetry | **T** ransaction costs | **L** iquidity creation | **R** isk sharing | **T** rust

"**I** **T**rust **L**enders, **R**arely **T**ellers."

Dimension	How a traditional bank does it	How FinTech does it
Information asymmetry	Relationship banking, collateral and manual underwriting fight **adverse selection** (before the contract) and **moral hazard** (after it).	Real-time alternative data, continuous transaction analytics and ML scoring; automated monitoring.
Transaction costs	Economies of scale from branches. Costs have been "**sticky**" at about **2% of transaction value since the 1880s**.	Cloud, APIs and smart contracts automate workflows, pushing marginal cost toward zero.

Liquidity creation	**Maturity transformation**: short-term deposits become long-term loans (e.g. 30-year mortgages).	Fractional assets, secondary loan markets, instant order matching, less balance-sheet load.
Risk sharing	Pools uncorrelated risks on a big balance sheet; guarantees to depositors.	Unbundles risk: fractional investing, auto-diversification, AI/big-data risk pricing.
Trust	Institutional brand, deposit insurance, physical branches.	Algorithmic and cryptographic trust: transparent code and verifiable ledgers.

2.3 Christensen's disruptive innovation



Sustaining innovation
Incremental upgrades that incumbents make to protect their best, high-margin customers. Over time they **over-serve** the market.

Disruptive innovation
Entrants with fewer resources start at the **low end or with non-consumers** using simpler, cheaper offers, then **march upmarket** until they are good enough for the mainstream.

ADDED FROM SLIDES · MCKINSEY DIAGNOSTIC: WHEN IS AN INCUMBENT VULNERABLE?

1. Core customers **cross-subsidise** unprofitable segments.
2. Customers must buy a whole **bundle** to get one feature.
3. Services aren't available **on demand (24/7)**.
4. User journeys fall **below modern digital standards**.

2.4 Pisano's innovation landscape map

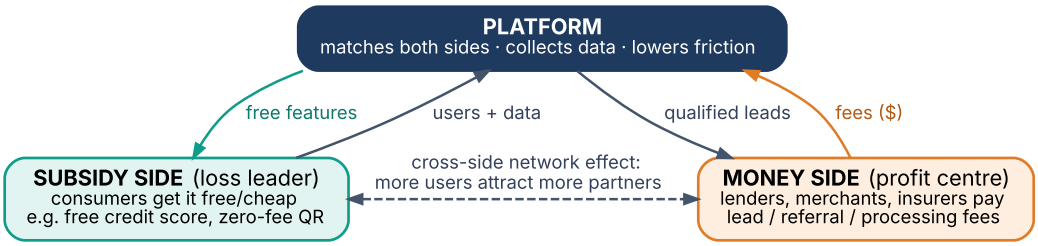
BUSINESS MODEL: NEW ↑	TECHNOLOGY: EXISTING ←	TECHNOLOGY: → NEW
	DISRUPTIVE Existing tech + <u>new</u> business model. *e.g. marketplace lending, zero-commission trading*	**ARCHITECTURAL** New tech + new business model. Hardest to execute. *e.g. DeFi protocols, smart contracts, DAOs*
EXISTING ↓	**ROUTINE** Existing tech + existing model = Christensen's *sustaining*. Source of most profit. *e.g. bank upgrades its mobile app*	**RADICAL** <u>New</u> tech (R&D) + existing model. *e.g. AI fraud / credit models inside a normal bank*

The pivot imperative: start-ups often launch with routine or radical ideas but must pivot to **disruptive or architectural** models to survive against well-funded incumbents.

> **EXAM TRAP**
>
> "Radical" ≠ "disruptive". **Radical = new technology** (business model unchanged). **Disruptive = new business model** (technology can be old). Zero-commission brokerage uses old tech with a new model, so it is **disruptive**.

2.5 One-sided vs multi-sided platforms

One-sided (legacy): the firm earns from one customer side via interest spreads, management fees or premiums. **Multi-sided platform (MSP):** connects two or more interdependent groups (borrowers/investors, buyers/merchants, consumers/developers).



Rochet & Tirole platform economics: subsidise one side, charge the other.

Same-side (direct) network effect
More users on one side make it better for that same side (e.g. more friends on a P2P payment app).

Cross-side (indirect) network effect
More users on one side attract the other side (more cardholders → more merchants accept the card).

Revenue profiles: agency vs principal

Type	Meaning	Who (King, 2020 data)
Agency-based (non-interest income)	Fees, commissions, transaction charges, **no balance-sheet risk**	Most FinTechs (Revolut, LendingClub, Wealthfront, Credit Karma): **90–100% agency**

Principal-based	Net interest income (spread), investment yields, **takes balance-sheet risk**	Banks, e.g. JPMorgan: **65% principal / 35% agency**

Exceptions: SoFi moved to holding loans (**31% principal**); ZhongAn (digital insurer) earns **96%** from balance-sheet underwriting.

Platform case studies

Credit Karma (MSP, acquired for ~$7B)
Subsidy side: consumers get free credit scores, monitoring and tax tools.
Money side: card issuers and lenders pay lead-generation and conversion fees when users apply for matched products. (Borrowell is the Canadian equivalent.)

Paytm & Ant Group (Alipay)
Scaled with **zero-fee QR payments** (subsidy side), then cross-sold merchant processing, micro-loans, wealth (e.g. Yu'e Bao) and insurance (money side).

2.6 Seven current trends

#	Trend	What to remember
1	**Digital banking: unbundling → rebundling**	Start-ups carved off profitable silos (payments, FX, micro-loans); neobanks (Nubank, Revolut, Chime) then **rebundled** them into all-in-one apps. Catalysts: PSD2 / UK Open Banking. Incumbents respond with spin-offs (Marcus by Goldman Sachs), M&A, VC arms, innovation hubs.
2	**Digital payments**	A **~$2 trillion** industry, **~40% of commercial bank revenue**; has minted **more unicorns** than any other vertical. Wise bypasses SWIFT; Stripe and Adyen turned acquiring into APIs.
3	**Alternative finance**	P2P / debt crowdfunding ≈ **90%** of alt-finance volume; equity ≈ **3%**; donation + reward ≈ **7%**. ML underwriting for thin-file borrowers.
4	**Robo-advisors**	Democratising **$100T** AUM; fees fell from 1–2% to **25–50 bps**; minimums from $250k+ to low; Robinhood made trades commission-free; shift to **hybrid** models.
5	**InsurTech**	The **$6T** insurance value chain; cloud-first carriers (ZhongAn), **telematics** and wearables for individual pricing, computer-vision claims.
6	**Crypto, DeFi, smart contracts**	Bitcoin (SHA-256 + PoW), Ethereum (Turing-complete smart contracts), DeFi (AMMs, DEXs, lending pools), **stablecoins** (USDT, USDC) pegged to fiat.
7	**TechFins & BigTech**	**FinTech** starts with finance and improves delivery; **TechFin** starts with a platform (e-commerce, social, hardware) and embeds finance. Ant / Tencent super-apps; Apple Pay / Apple Card (with Goldman & Mastercard); Amazon merchant lending; Meta and Google pulled back after regulatory push-back. Moat: users, data, network effects.

Where it is heading: from **disruption to collaboration** (FinTechs bring front-end agility and software; incumbents bring compliance, insured deposits and scale), **embedded finance** inside non-financial apps, and **data** as the main intermediation asset.

2.7 Exam practice

10 marks **Evaluate a zero-commission brokerage using Pisano's map and intermediation theory. Explain transaction costs, liquidity creation and cross-side network effects.** CO1, CO2

1 · Pisano positioning: Disruptive

It uses **existing technology** (mobile apps, order routing, clearing houses) with a **new business model**: no $5–10 per-trade commission, monetised in the back end instead.

2 · Intermediation dimensions

- **Transaction costs:** automated onboarding and clearing push costs toward zero, so small trades become economical.
- **Liquidity creation:** fractional shares (e.g. $5 of a $2,000 stock) pool retail money and raise trading volume.

3 · MSP dynamics

- **Subsidy side:** retail traders get free trades and free data.
- **Money side:** market makers pay **Payment for Order Flow (PFOF)**, plus the platform earns interest on uninvested cash and securities lending.
- **Cross-side effect:** more retail users → more order flow → more market makers willing to pay.

4 · Trade-offs

Strengths: fast user growth, scalable monetisation. **Risks:** regulatory scrutiny of PFOF and execution quality; gamification may encourage over-trading (principal–agent conflict).

5 marks **How does information asymmetry show up as adverse selection and moral hazard in credit markets, and how does alternative data help?** CO1, CO2

Borrowers know more about their own repayment ability than lenders do.

- **Adverse selection (before the loan):** risky borrowers apply most eagerly. If the lender can't tell them apart it raises rates for everyone, which drives **safe** borrowers away.
- **Moral hazard (after the loan):** once funded, the borrower may take more risk or misuse funds because the lender bears the loss.

FinTech lenders reduce **adverse selection** with alternative data (utility payments, e-commerce cash flows, bank-account API data) that sees beyond a static credit score. They reduce **moral hazard** with continuous open-banking monitoring, automatic repayment deductions from daily POS sales, and dynamic credit limits.

60-SECOND RECALL · UNIT 2

- ☐ Draw the pentagon and say how FinTech changes each of the five dimensions.
- ☐ Sketch Christensen's chart: foothold → upmarket march → over-serving incumbent.
- ☐ Draw Pisano's 2×2 and place: P2P lending, DeFi, bank app upgrade, AI credit model in a bank.
- ☐ Subsidy side vs money side, with Credit Karma as the example.
- ☐ JPMorgan 65/35 vs FinTech 90–100% agency. Which two FinTechs are exceptions?
- ☐ FinTech vs TechFin, with one example each.

03 Artificial Intelligence & Machine Learning in Finance

TIER 2 · CORE

Course outcomes: CO2 (role of AI, big data) · CO4 (ethics, regulation)

In one line: ML replaces hand-written "if-then" rules with patterns learned from **big data**, turning weeks of manual underwriting, advice and claims into **minutes or seconds**. The price is **bias** and the **"black box"** problem, which **explainable AI** has to solve.

3.1 What AI and ML are, and why now

AI = machines doing cognitive tasks that mimic human intelligence (reasoning, planning, perception, language). **ML** = a branch of AI where algorithms **learn decision rules from data** instead of programmers writing deterministic rules.

History
1950s–60s academic start → "**AI winter**" (not enough hardware or data) → **2000s renaissance**.

Three catalysts of the renaissance
① Massive compute (GPUs, clusters) ② cheap elastic cloud storage (**AWS 2006**) ③ an explosion of digital financial data.

3.2 The three ML paradigms

Paradigm	How it works	FinTech uses
Supervised known inputs → known target	Trains on **labelled** history, mapping X → Y (classification / regression, e.g. logistic regression).	Credit scoring, default prediction, mortgage PD, churn, **known** fraud patterns.
Unsupervised find hidden structure	**Unlabelled** data (X only); clustering such as k-means or hierarchical groups items by similarity.	**AML anomaly detection**, new fraud rings, customer segmentation.
Reinforcement agent ↔ environment	An agent acts, gets a reward or penalty, and learns by trial and error to maximise cumulative reward.	Executing large trade blocks with minimal market impact, order routing, adaptive robo rebalancing.

MNEMONIC · "S-U-R"

S upervised = Scoring (labels) | **U** nsupervised = Unusual patterns (AML) | **R** einforcement = Rewards (trading)

3.3 Big data: the fuel

"**Data is the new oil**" (Clive Humby, 2006): raw data has latent value but must be refined and fed into models. Over **90%** of the world's data was created in the last few years.

Structured data
Relational databases, account statements, bureau scores, payment amounts.

Unstructured data
Browsing trails, geolocation, POS timestamps, social activity, text and voice.

Infrastructure: **Apache Hadoop** and **Apache Spark** (distributed storage and processing), data pipelines and dashboards for data engineers and scientists. **NLP** turns text and audio into inputs: sentiment from earnings calls and news, chatbots, and extracting data from invoices and tax filings.

3.4 ML credit underwriting

The legacy gap: bank underwriting is slow and paper-heavy and penalises **thin-file** people and SMEs with little bureau history. **Alternative-data features:** bank cash inflows and expense volatility, POS inventory turnover, supplier payment reliability, device and digital-footprint data, e-commerce history.

1 · Data in bank APIs, POS cash flow, utility bills, invoices → **2 · Features** NLP on text, revenue volatility, normalising → **3 · ML model** e.g. gradient-boosted trees → probability of default → **4 · XAI audit** SHAP / LIME, bias check, reason codes → **5 · Decision** approve, limit, price: in **minutes**, not 2–4 weeks

ADDED FROM SLIDES · KING'S REALITY CHECK (LIKELY EXAM POINT)

Academic studies show that although platforms advertise "proprietary alternative-data" scoring, actual marketplace credit decisions are **still heavily correlated with conventional hard bureau data**. Behavioural quirks also matter: retail lenders avoid borrowers who make **spelling errors** or write unusually **long or short** descriptions.

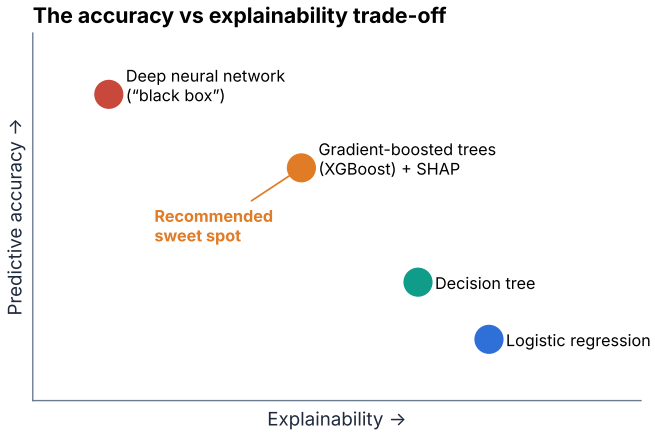
3.5 Ethics: bias, black boxes and other risks

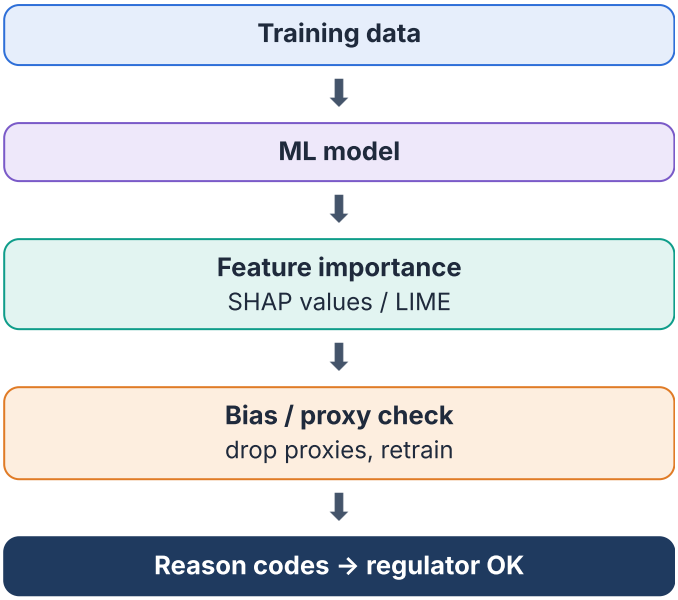
Algorithmic bias → "redlining"
Models trained on historic bank data absorb human bias. **Proxies** (postcode, school, shopping categories) stand in for protected traits, producing higher rates or rejections for protected groups.

The black-box problem
Deep neural nets have millions of non-linear parameters and can't give human-readable reasons, which clashes with laws requiring **adverse-action notices** (e.g. US Equal Credit Opportunity Act).

Principal–agent engagement loops
Recommendation engines tuned for platform commission can push users to high-margin credit or speculative trading.

Model drift & systemic fragility
Models trained in calm markets fail in crises and can trigger correlated sell-offs and liquidity crunches.





The XAI audit loop

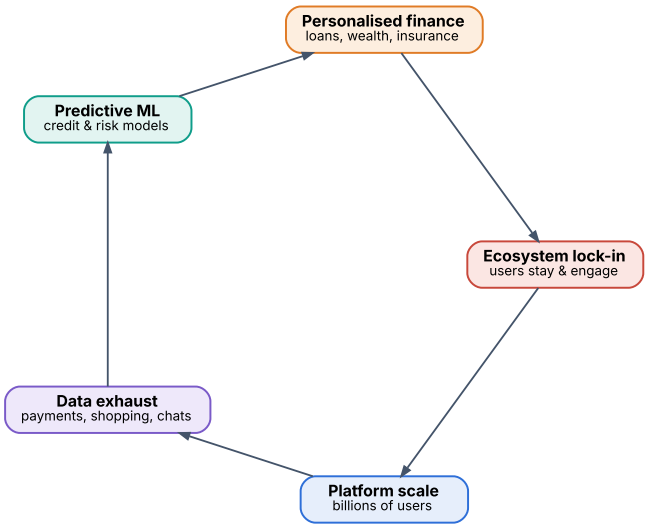
SHAP (SHapley Additive exPlanations) measures each feature's contribution to a decision. **LIME** (Local Interpretable Model-agnostic Explanations) approximates one decision with a simple local model.

3.6 AI across the business lines

Wealth: robo-advisor algorithms

Automated **risk profiling** (questionnaire → utility curve) · **mean-variance optimisation** across low-cost ETFs · **continuous rebalancing** when weights drift · **tax-loss harvesting** (sell losers to offset gains, buy a correlated replacement). Fees fell from **1–2%** with **$250k+** minimums to **0.25–0.50%** with **$500–$5,000** minimums. High customer-acquisition cost pushed robos toward **hybrid** models.

BigTech data flywheel



Stand-alone FinTechs pay $500–$1,000 CAC per user; BigTech already has billions of users.

Ant / Tencent: use payment velocity to approve merchant micro-credit instantly. **Amazon Lending:** uses inventory velocity, ratings and returns to pre-approve working capital. **Apple:** FaceID, Secure Enclave and app history for payments and co-branded credit.

Case studies

Lendified & Judi.ai: SME underwriting

Toronto, **2015**, founded by ex-bankers Troy Wright & Kevin Clark. **Who:** SMEs needing **$5k–$150k** working capital. **Pain:** weeks of paper-based bank review, or predatory merchant cash advances.

Solution: API pull of business bank data → ML cash-flow forecasting → application in **<10 min**, decision in minutes, money in **48 h**. **Pivot:** balance-sheet lender *plus* licensing **Judi.ai as B2B SaaS** to banks and credit unions, cutting their adjudication cost by up to **90%**.

Lemonade: conversational AI insurer

US renters, home and pet insurer for digital-native millennials. **Pain:** jargon, long forms, adversarial slow claims.

AI Maya (NLP onboarding bot) binds a policy in about **2 minutes**. **AI Jim** (claims bot) checks video and receipts, runs anti-fraud checks and pays in as little as **3 seconds**. A continuous ML loop learns from claims data.

ZhongAn: cloud-native micro-insurance

China's first **digital-only insurer**, founded **2013** by the "**Three Mas**" (Jack Ma of Alibaba, Pony Ma of Tencent, Mingzhe Ma of Ping An); listed in Hong Kong in **2017**. **Pain:** tiny-premium policies (shipping-return, flight-delay) are too costly to administer manually.

100% cloud, prices risk per user per transaction, issues **millions of policies a day** at peak shopping events, automated claims.

Wealthfront & Wealthsimple

Wealthfront (Silicon Valley) and Wealthsimple (Canada, **2014**, Mike Katchen). **Who:** young professionals and **HENRYs**. **Pain:** 1–2% fees and $250k+ minimums.

Solution: risk questionnaire → passive ETF allocation → auto-rebalancing → algorithmic tax-loss harvesting, with no human advisor.

3.7 Exam practice

5 marks **Explain how unsupervised ML is applied to AML monitoring, contrasting it with supervised learning.** CO2, CO4

Laundering methods change quickly, so **labelled history is incomplete**. **Unsupervised** models take **unlabelled** transactions and use clustering to find hidden groups, unusual links between accounts and outliers from normal behaviour, without predefined rules. That suits **new** schemes.

Supervised models need labelled data (X → Y: each past transaction marked legitimate or fraudulent). They are excellent at catching **known** patterns but blind to schemes that have no labels yet. In practice banks use both: supervised for known fraud, unsupervised to discover new rings.

10 marks **A bank wants to replace its credit scorecard with a deep neural network. Evaluate the accuracy vs explainability trade-off under fair-lending rules.** CO2, CO4

1 · The trade-off

- **Accuracy:** DNNs capture non-linear patterns across thousands of alternative-data variables, cutting NPLs and sharpening risk pricing.
- **Explainability:** hidden layers make it hard to trace why a given applicant was rejected (the black box).

2 · Regulatory constraint

Fair-lending law (e.g. US ECOA) requires **adverse-action notices** with specific reasons, and lenders must show they don't use protected traits or **proxies** (e.g. postcode correlating with race) that cause **disparate impact**.

3 · Solution: an XAI stack

1. **SHAP**: quantifies each feature's contribution.
2. **LIME**: approximates individual decisions with a simple model.
3. **Global feature-importance audits**: confirm proxies are removed.

4 · Recommendation

Use **gradient-boosted trees (e.g. XGBoost) with SHAP** rather than an unconstrained DNN. That keeps most of the accuracy gain while preserving audit trails and compliance. (Link to the lecture's discussion question: even a model that is "20% more accurate" should not be allowed if it can't explain rejections.)

60-SECOND RECALL · UNIT 3

- ☐ The three catalysts of the 2000s AI renaissance.
- ☐ S-U-R: match each to a FinTech use case.
- ☐ Draw the five-step ML underwriting pipeline.
- ☐ King's reality check: what still drives marketplace credit decisions?
- ☐ Four AI risks: bias/redlining, black box, engagement loops, model drift.
- ☐ Lemonade's two bots and their times; ZhongAn's founders; Lendified's numbers.

04 Blockchain Technology & Cryptocurrencies

TIER 2 · CORE

King, Chapter 5 · Course outcomes: CO2 (blockchain) · CO4 (risks, regulation)

In one line: Bitcoin solves the **double-spend problem without a bank** by combining **SHA-256 hashing + public-key cryptography + a shared ledger + Proof-of-Work**. It is secure but **slow (2–4 TPS)**, energy-hungry and, as an asset, **very volatile**.

4.1 The double-spend problem and the birth of Bitcoin

A digital token is just bits, so it can be copied. Without a referee you could send the **same coin to Seller A and Seller B**. Traditionally a **trusted central ledger** (a bank or the mint) prevents this.

The 2008 breakthrough

October 2008: "Satoshi Nakamoto" publishes *Bitcoin: A Peer-to-Peer Electronic Cash System*, describing purely P2P online payments with no financial institution. It came out of the GFC and distrust of bailouts.

Genesis Block (Block 0), 3 Jan 2009

Embedded headline: *"The Times 03/Jan/2009 Chancellor on brink of second bailout for banks."*

4.2 Cryptographic foundations

SHA-256 hashing

Takes **any input** (one character or a whole ledger) and maps it to a **fixed 256-bit** (32-byte) output, written as **64 hex characters**. "Crypto" comes from the Greek *kryptos*, "hidden".

Deterministic
Same input always gives the same hash.

One-way
Hash = H(x) is easy; recovering x from the hash is practically impossible (pre-image resistance).

Avalanche effect
Change one bit → a completely different hash.

Collision-resistant
Two different inputs giving the same hash is infeasible.

```
SHA-256("fintech") = 110599ccb008c6746985524be4ba99ff588fed6bc5c4d65d2a15b6a92eda16a7
SHA-256("Fintech") = 2dd47261bf9305896542a63678477884cba21c4b70fab4b8fafb69be2672608d
                     ↑ one capital letter changed → totally different output (avalanche
effect)
```

MNEMONIC · HASH PROPERTIES "F-DOAC"

F ixed 256-bit length | **D** eterministic | **O** ne-way | **A** valanche | **C** ollision-resistant

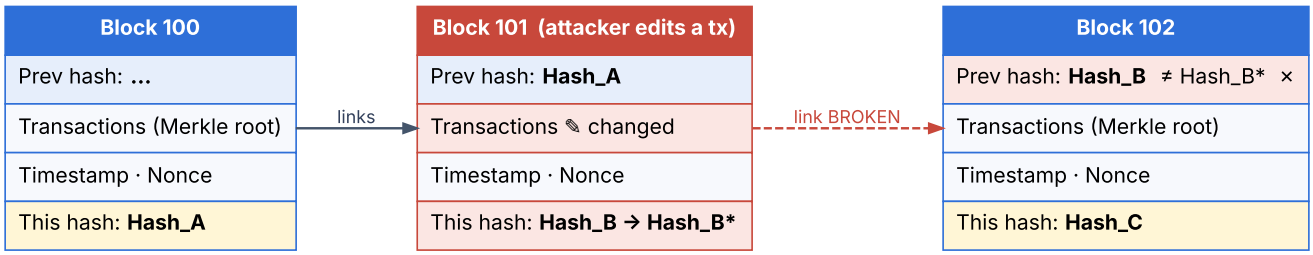
Your first draft used "D-O-D-A" with "Difficulty-adjusted", but difficulty is a **mining** rule, not a hash property. Keep them separate in answers.

Public-key (asymmetric) cryptography

- **Private key**: secret; signs (authorises) transfers and unlocks your coins. Lose it and the coins are gone.
- **Public key / address**: derived from the private key and shared openly (Satoshi's genesis address: 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa).
- **Pseudonymity**: addresses hide real identities until coins are off-ramped into fiat.

4.3 How blocks are chained, and why tampering fails

A blockchain is a digital, **append-only**, distributed ledger. Transactions are batched into **blocks**; each block header stores the **hash of the previous block header**.



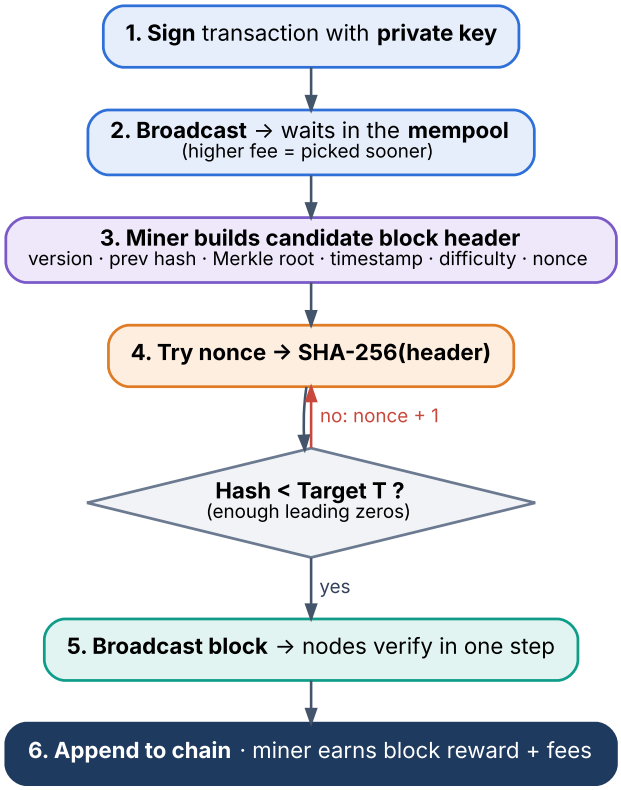
Edit one old transaction → its hash changes (avalanche) → the next block's "prev hash" no longer matches → every later block is invalid.

To get away with it, an attacker must **re-mine that block and every later block faster than the rest of the network combined**, which needs majority hash power (a **51% attack**).

ADDED FROM SLIDES · THE 9 FEATURES OF THE BITCOIN BLOCKCHAIN (KING)

1 Form	100% digital record	**6 Consensus**	Proof-of-Work mining
2 Contents	Chronological BTC transfers between addresses	**7 Throughput**	1 MB block every ~10 min → **2–4 TPS**
3 Security	SHA-256 + public-key cryptography	**8 Interoperability**	Closed: verifies BTC only, no built-in bridges
4 Transparency	Fully public; anyone can inspect	**9 Governance**	Open-source (Bitcoin Core), node consensus
5 Access	Permissionless: join without approval		

4.4 Proof-of-Work mining



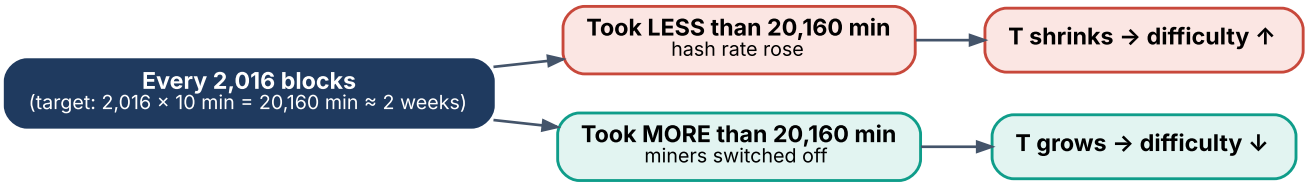
Miners hash the candidate **block header** (version, previous hash, **Merkle root** of transactions, timestamp, difficulty bits) plus a **nonce** ("number used once").

```
Winning condition:  SHA-256(header + nonce) < T
T = target threshold. A smaller number has more leading zeros:
  000000000000000002a4b87…  ✔ valid
```

The output is pseudo-random, so the nonce **can't be calculated**; it has to be **brute-force guessed**. Finding it is hard but **checking it takes one hash**. That asymmetry is the whole trick.

The winner earns the **block reward** plus **transaction fees**.

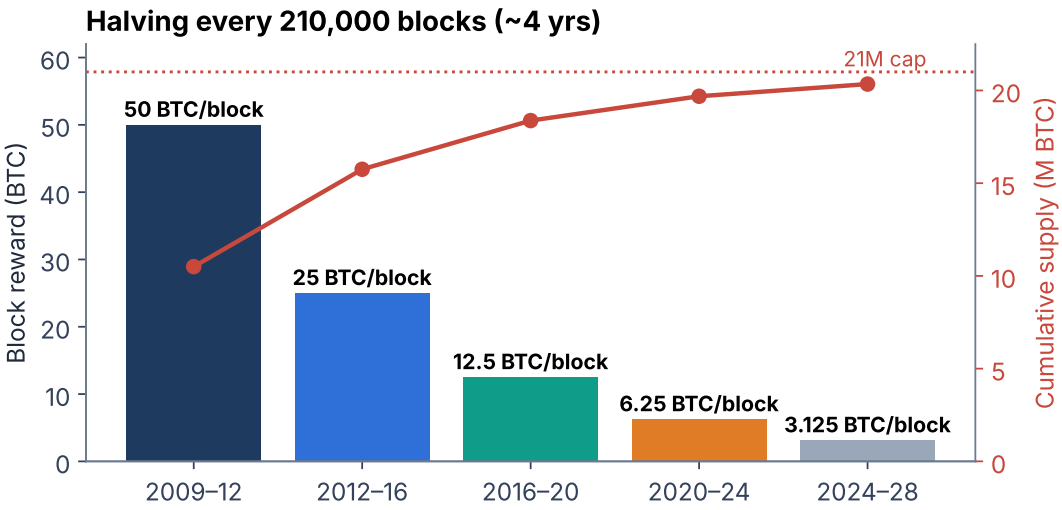
Difficulty adjustment (keeps blocks at ~10 minutes)




```
Difficulty  D = T_max / T            (T and D move in opposite directions)
T_new = T_current × (Actual minutes for last 2,016 blocks / 20,160)
  ratio < 1 (too fast)  → T shrinks → harder
  ratio > 1 (too slow)  → T grows   → easier
```

Energy: as hash rates reached trillions of hashes per second, Bitcoin used about **73 TWh a year (≈ Austria)**. **Proof-of-Stake** (Ethereum's "The Merge") removes the mining race and cuts energy use by **99.95%**.

Fixed supply and the halving



Era	Blocks	Reward / block	BTC issued
2009–2012	0 – 209,999	50	10,500,000
2012–2016	210,000 – 419,999	25	5,250,000
2016–2020	420,000 – 629,999	12.5	2,625,000
2020–2024	630,000 – 839,999	6.25	1,312,500
Geometric series → total converges to			**21,000,000 cap**

Throughput and the mempool: the bus analogy

Bitcoin is a bus that leaves every **10 minutes (600 s)** with **1 MB** of seats. Each transaction (passenger) takes **250–500 bytes**, so about **1,200–2,400** fit (call it 1,500).

```
TPS ≈ 1,500 / 600 s ≈ 2.5 transactions per second
Visa ≈ 24,000 TPS
```

The mempool is the crowd at the bus stop: unconfirmed transactions waiting to be included. Passengers **bid with fees**, and the miner (driver) picks the **highest fees first**. Low-fee transactions wait. During congestion, fees spike.

4.5 Consensus: Proof-of-Work vs Proof-of-Stake

Criterion	Proof-of-Work (PoW)	Proof-of-Stake (PoS)
Who adds the block	Miner who first solves the SHA-256 nonce puzzle	Validator chosen according to the tokens they **lock up (stake)**
Resource	Hardware (ASICs) + electricity (~73 TWh/yr)	Capital; ~99.95% less energy
Security	Cost of energy; **51% hash-power attack**	Financial penalty: **slashing** of stake for double-signing; 51% *capital* attack
Throughput	Low (2–4 TPS on Bitcoin; often quoted up to ~7)	Higher, with faster confirmation

Finality	Probabilistic (wait ~6 confirmations)	Deterministic via validator voting
Weak spot	Fee spikes and slow confirmation under congestion	"Nothing at stake": voting on several forks at no cost, solved by slashing

Other designs: **Ripple** uses a Byzantine-fault-tolerant consensus run by approved validator nodes (see the case below).

4.6 Forks and the "free money" paradox

Soft fork
Backward-compatible upgrade; old nodes still accept the new blocks.

Hard fork
Not backward-compatible; the chain **splits permanently**. Both chains share the same history up to the split block.

THE BITCOIN CASH HARD FORK, 1 AUG 2017 (BLOCK 478,558)

Dispute over block size: Core kept **1 MB**, dissenters went to **8 MB**. Every BTC holder got **1 BCH free**.

```
Theory (pizza logic):   P_pre-fork = P_BTC,after + P_BCH   → no new value
Reality:   before 1 BTC = $2,746
           after  1 BTC = $2,741   +   1 BCH = $380   =  $3,121
           holders "gained" $375 per coin  (+13.6% overnight)
```

Lesson: crypto markets were not behaving rationally. Hype created paper wealth from nothing, which points to inefficient, speculation-driven pricing.

4.7 Crypto as an asset

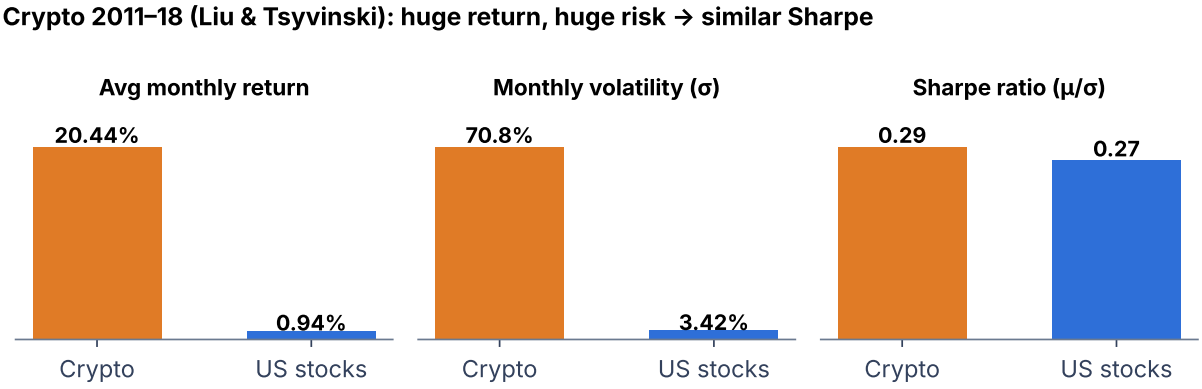
ICOs (Initial Coin Offerings)

A crowdsale of pre-mined tokens straight to the public via a white paper, with **no investment-bank underwriting**.

Year	No. of ICOs	Raised ($M)
2014	7	30.4
2015	7	8.6
2016	43	256.4
2017	343	5,482.0
2018 (Jan–Jul)	460	14,295.4
Total	**860**	**$20.07 billion**

Mega-deals: **EOS $4.2B** (June 2018), **Telegram $1.7B** (March 2018). In mid-2018 the US SEC applied the **1946 Howey Test**: *investment of money + common enterprise + expectation of profit + from the efforts of others = a security*. That effectively shut down public ICOs.

Risk and return



Crypto returns are **not** explained by mining cost or difficulty. They are predicted by **adoption** (active addresses, wallet downloads), **attention** (Google searches) and **momentum**.

① Market factor ("the tide")
When Bitcoin moves, everything moves. BTC co-movement explains about **80%** of altcoin price variance.

② Size factor (SMB)
Smaller, cheaper coins earn higher % returns as compensation for higher risk.

③ Momentum factor
Winners over the last **1–4 weeks** tend to keep winning for the next 1–4 weeks.

Three factors explain about 90% of the variation out of the 24 variables tested. **Cross-exchange arbitrage:** the same coin can trade at different prices on different exchanges. Gaps persist because of capital controls, withdrawal limits and counterparty risk, which is why the CME's Bitcoin futures did not close them immediately (context added to answer the lecture's discussion question).

4.8 Case studies

Ripple (XRP): interbank settlement

- Founded **2011** as OpenCoin (Jed McCaleb, David Schwartz, Arthur Britto; Chris Larsen joined in 2012; renamed Ripple in 2013). Goal: a "free PayPal for the world" fixing slow correspondent banking.
- Replaced PoW with the **Ripple Protocol Consensus Algorithm (RPCA)**: approved validators on a Unique Node List confirm in seconds. **100 billion XRP pre-mined**; no new coins.
- Pivoted from retail users to **banks**: RippleNet and **On-Demand Liquidity** use XRP as a bridge currency between fiat pairs.
- Faced an **SEC lawsuit** claiming the pre-mined XRP sales were unregistered securities.

FTX: structural collapse (Nov 2022)

- Founded **2019** by Sam Bankman-Fried; became a top global exchange.
- **Conflict of interest:** FTX and its trading arm **Alameda Research**. FTX created the **FTT** token; Alameda held illiquid FTT as collateral to borrow **billions of customer deposits**.
- Binance said it was selling FTT → a classic **bank run** → FTT crashed → about an **$8B shortfall** → Chapter 11.
- **Lesson:** not a failure of blockchain technology but of **centralised governance, principal–agent moral hazard and fraud**.

ADDED FROM SLIDES · RISKS FACING "HODLERS" (KING TABLE 5.6)

"HODL" comes from a 2013 forum typo ("I AM HODLING"), later read as "Hold On for Dear Life". Risks: **Price** (>70% annual volatility, crypto winters) · **Counterparty** (exchange failures such as Mt. Gox, QuadrigaCX, FTX) · **Settlement** (irreversible; a lost key means lost coins) · **Operational / cyber** (smart-contract bugs, phishing, future quantum threat to SHA-256) · **Governance / fork** · **Legal / regulatory** (mixers, tax, licensing, stablecoin reserves).

RELATED TERMS TO DEFINE IN ONE LINE

DLT: a ledger shared across nodes with no central owner. **Smart contract**: self-executing "if A then B" code. **dApp**: an app running on a blockchain. **DeFi**: lending and trading via smart contracts. **AMM**: an automated market maker pricing assets with a formula ($x \cdot y = k$). **DEX**: a non-custodial exchange. **TVL**: total value locked in a DeFi protocol. **NFT**: a unique token proving ownership. **Stablecoin**: a token pegged to fiat (USDT, USDC). **CBDC**: central-bank-issued digital legal tender.

4.9 Exam practice

5 marks **Explain how a block hash links to prior blocks, and why altering a historical transaction invalidates the chain.** CO2

Each block header contains the **SHA-256 hash of the previous block header** along with the Merkle root of its own transactions, which creates a back-linked chain. If an attacker changes a transaction in Block 101, the **avalanche effect** changes Block 101's hash from Hash_B to Hash_B*. Block 102 still stores Hash_B as its "previous hash", so the link breaks, and every later block becomes invalid. Nodes reject the altered chain unless the attacker **re-mines Block 101 and all later blocks faster than the honest network**, which requires over 50% of hash power (a 51% attack). That is economically prohibitive.

10 marks **A global payment network is choosing between PoW and PoS for high-volume settlement. Compare consensus, energy, throughput, finality and attack vectors, and recommend.** CO2, CO4

Use the PoW vs PoS table in 4.5 as the backbone of the answer, then add:

- **Under congestion:** PoW fees spike as users bid for limited block space, and fixed ~10-min blocks plus probabilistic finality cap throughput. PoS handles parallel validation and finalises deterministically, which suits payment SLAs.
- **Attack vectors:** PoW = 51% hash-power attack to rewrite recent history and double-spend. PoS = "nothing at stake" (validating several forks at no cost), countered by **slashing** staked collateral.
- **Energy / ESG:** PoW ≈ 73 TWh/yr (Austria); PoS cuts this by ~99.95%.

Recommendation: PoS, for high TPS, fast deterministic finality, low energy and economic (slashing) security. Note the trade-off: stake concentration can centralise control.

60-SECOND RECALL · UNIT 4

- ☐ Double-spend problem in one sentence; date of the white paper and the Genesis Block.
- ☐ F-DOAC hash properties, with the fintech/Fintech example.
- ☐ PoW condition (hash < T); D = T_max/T; adjustment every 2,016 blocks / 20,160 min.
- ☐ Halving: 50 → 25 → 12.5 → 6.25; 210,000 blocks; 21M cap.
- ☐ Bus analogy: 1 MB, 10 min, ~2.5 TPS vs Visa 24,000.
- ☐ BCH fork: $2,746 → $2,741 + $380 = $3,121 (+13.6%).
- ☐ ICO total $20.07B / 860 ICOs; Howey Test 1946. Sharpe 0.29 vs 0.27.
- ☐ Ripple: RPCA, 100B pre-mined, SEC suit. FTX: FTT, Alameda, $8B hole.

05 Payment Systems & Digital Wallets

King, Chapter 9 (plus Ch. 3–4 valuation) · Course outcomes: CO3 (digital payments) · CO1

TIER 1 · HIGH WEIGHT

In one line: Card payments run through a **4-party network** that takes about **2.6%** of every sale, with the **issuer taking the biggest slice**. FinTechs attack this by removing layers (Wise, M-PESA, A2A rails, QR) or turning acquiring into code (Stripe, Adyen). Pure processing earns only **10–20 bps**, so winners cross-sell.

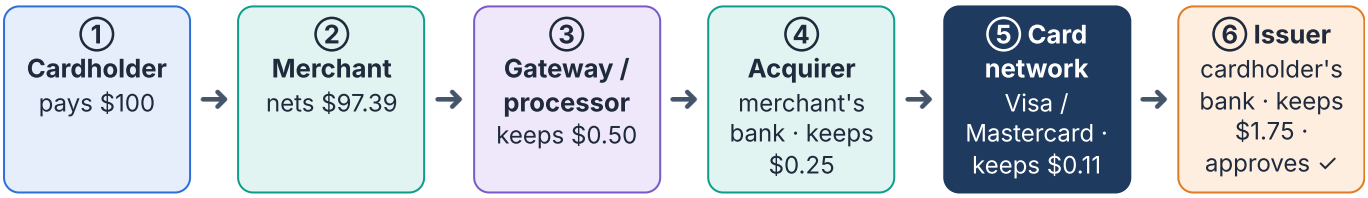
5.1 The $2 trillion value pool

KEY NUMBERS

Payments ≈ **US$1.9 trillion** of global revenue (McKinsey 2020), forecast above **$2.5T by 2025**, and about **40% of all commercial-bank revenue**. High frequency: King's benchmark is that 47M Chase customers average **5 ATM visits, 6 wallet payments, 32 debit taps and 21 credit-card transactions a month**. Payments has produced **more unicorns** than any other FinTech vertical.

Disruption playbook: anchor on **one** high-friction use case (P2P transfers, mobile POS readers, transparent FX), then cross-sell banking, lending and software.

5.2 The players and the transaction flow



Authorisation request travels left → right; approval comes back right → left; money settles later (Step 2).

- **Cardholder**: pays by plastic, NFC tap or web checkout.
- **Issuing bank**: issued the card; checks the balance or credit and **bears default risk**.
- **Merchant**: sells the goods. **Acquirer** (merchant's bank): holds the merchant's account.
- **Card network / scheme**: Visa, Mastercard, UnionPay; routes messages and clears. It **doesn't hold the money**.

Four-party scheme

Issuer and acquirer are separate banks (Visa, Mastercard).

Three-party scheme

The network is **both** issuer and acquirer (traditional American Express).

Two-step settlement

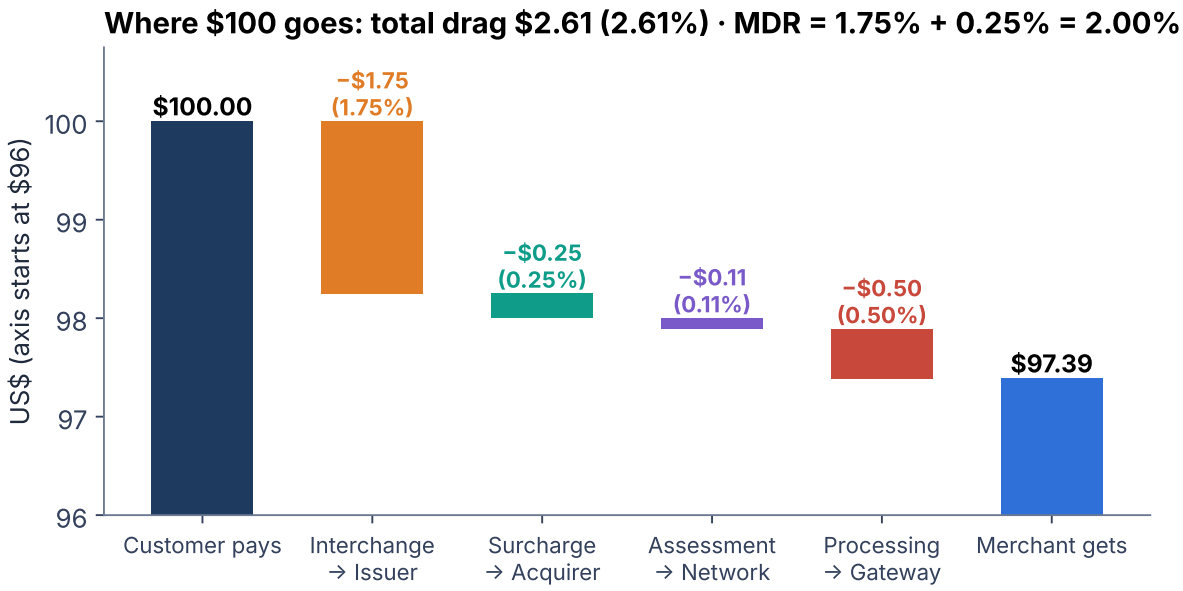
Step 1 · Authorisation
seconds
terminal → gateway → acquirer → network → issuer → "approved"

→

Step 2 · Clearing & settlement
1–2 days, batched end-of-day
funds move issuer → acquirer → merchant

This is why your bank statement shows a "transaction date" and a later "posting date".

5.3 The fee waterfall (King Table 9.1): learn this cold



Line item on a $100 card purchase	Amount	Rate	Who gets it · why
Customer charge	$100.00	–	Gross sale
Interchange fee	−$1.75	1.75%	**Issuer**: default risk, fraud, rewards, credit float
Merchant surcharge	−$0.25	0.25%	**Acquirer**: onboarding, terminals, servicing
= Merchant Discount Rate (MDR)	*−$2.00*	*2.00%*	*interchange + surcharge*
Network assessment	−$0.11	0.11%	**Visa / Mastercard**: routing and switching
Processing fee	−$0.50	0.50%	**Gateway / processor**: software, authorisation
Merchant receives	**$97.39**	**97.39%**	**Total intermediation drag = $2.61 (2.61%)**

```
MDR          = Interchange + Merchant surcharge          = 1.75% + 0.25% = 2.00%
Total drag   = MDR + Network assessment + Processing     = 2.00% + 0.11% + 0.50% = 2.61%
Merchant net = Gross × (1 − total drag)                  = $100 × 0.9739 = $97.39
Issuer share of total fees = 1.75 / 2.61 ≈ 67%  (the biggest slice)
```

EXAM TRAP · TWO WAYS MDR IS DEFINED

King's table defines **MDR = interchange + acquirer surcharge**, with network and gateway fees on top. Some questions instead give an **all-in MDR** that already includes the network fee, so the acquirer keeps the **residual**. Read the question, say which definition you're using, and show the arithmetic. Variant B below shows the all-in method.

5.4 Wallets, rails and processors

Type	How it works	Example
Closed-loop wallet	Money loaded into one brand's system; spendable only there	Starbucks app; originally Octopus

Open-loop wallet	**Tokenises** your real card for contactless and online payments over card rails	Apple Pay, Google Pay
QR-code networks	Cheap merchant acceptance with no terminal; common in emerging markets	Paytm, Alipay, UPI QR
Real-time A2A rails	Account-to-account instant settlement that **bypasses card networks**, at lower cost	National instant-payment systems (e.g. UPI)
Full-stack developer APIs	Gateway + acquiring + fraud + multi-currency payouts in one API	Stripe, Adyen

```
Platform revenue = Gross Merchant Volume (GMV) × take rate
Pure processing take rate ≈ 10–20 bps  → must add hardware/SaaS subscriptions and lending to profit
```

5.5 Case studies

① Paytm: multi-sided platform, unit-economics problem

Founded **2010**, New Delhi, by **Vijay Shekhar Sharma** ("Pay Through Mobile"); backed by Ant Group, Alibaba and SoftBank. Scale: **337M consumers, 22M merchants**.

Monetisation: merchant devices (QR, **Soundbox**, Android POS) at **$1.20–$3.10 per device per month**; consumer loans and BNPL via partner banks: **2.5–3.5% origination** + **0.5–1.5% collection** fee.

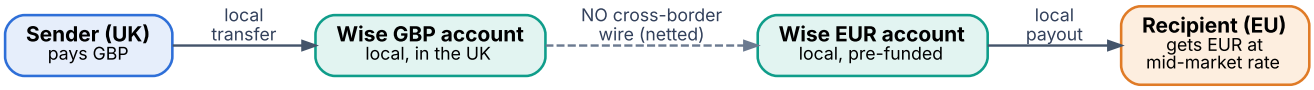
IPO Nov 2021: India's largest at the time, **$2.5B at an $18.5B valuation**; fell **27% on day one** and **66% within a year**.

② Wise (TransferWise): P2P FX netting

London, **2011**, Estonians **Taavet Hinrikus** (Skype employee #1, paid in EUR) and **Kristo Käärmann** (Deloitte, paid in GBP, EUR mortgage). They simply swapped money locally at the Reuters mid-market rate, with no wire fees and no hidden mark-up.

```
TAM = $20T global cross-border payments
SAM = $2T individual transfers × 4%
digital nomads = $80B
SOM = 50% share × $80B × 0.5% fee = $200M revenue
```

Expanded to **175 countries, 10 holding currencies**, forcing banks to cut hidden mark-ups.



Why Wise is cheap and fast: money never crosses the border; it is matched against opposite flows and paid out locally.

Paytm financials (King Table 9.2, INR millions)

Line	FY2020	FY2021	FY2022	% rev 2022
Total revenue	32,808	28,024	49,742	100%
Payment processing charges	22,659	19,168	27,538	55%
Sales & marketing	13,971	5,325	8,554	17%
Other operating expenses	27,314	23,234	39,482	79%
Total operating expenses	63,944	47,727	75,574	152%
Net operating loss	**−28,422 (−87%)**	**−16,961 (−61%)**	**−23,929**	**−48%**

Read it as: processing charges alone eat **55–69% of revenue** (paid to card networks and banks) and total opex stays **above 100% of revenue**. Payments alone can't make money on thin take rates without lending scale. FY2022 balance sheet: assets INR 179,916M, liabilities 38,400M (21%), equity 141,516M (79%).

③ M-PESA (Kenya)

Developed **2005**, launched **March 2007** by **Safaricom** (Vodafone affiliate) with the UK's **DFID**. "M" = mobile, "Pesa" = Swahili for money. Runs on the **SIM Toolkit + encrypted SMS** on basic 2G phones with no internet. **Agents** (airtime shops) act as human ATMs for cash-in and cash-out. It now dominates Kenyan retail payments and is the global inclusion case study.

④ Octopus Card (Hong Kong)

Launched **1997** by the MTR with bus and ferry operators; **contactless RFID**. **2000**: HKMA special deposit-taking licence turned it into general stored value; **2004**: parking meters. **33M cards** for 7.5M people (4.4 each); used by **98%** of 15–64 year-olds; **140,000 merchants**; **15M transactions a day**.

⑤ Stripe vs Adyen

Stripe (2010, Patrick & John Collison): developer-first; card acceptance in **7 lines of code**; **$95B** private valuation (Series H); 100+ currencies. **Adyen** (Amsterdam, 2006): one proprietary end-to-end omnichannel platform replacing a "spaghetti" of gateways. Valuation (King Table 4.6): market cap **$44B**, P/E **75.5×** LTM / **50.2×** NTM, EV/Sales **20.3×**, EV/EBITDA **33.1×** (NTM). Unified acquiring earns premium multiples.

ZhongAn (also used here): embeds micro-insurance into payment checkout (shipping-return, delay cover) in real time.

5.6 Exam practice

10 marks **Calculation: a customer buys an item for $100 by credit card. Interchange is 1.75%, the acquirer's surcharge 0.25%, the network assessment 0.11% and the gateway processing fee 0.50%. Find the MDR, each party's share and the merchant's net settlement, and comment on the economics.** CO3

```
MDR        = 1.75% + 0.25% = 2.00% → $2.00
Issuer     = $100 × 1.75% = $1.75
Acquirer   = $100 × 0.25% = $0.25
Network    = $100 × 0.11% = $0.11
Gateway    = $100 × 0.50% = $0.50
Total fees = $2.61 (2.61%)  →  Merchant net = $100 − $2.61 = $97.39
```

Economics: the issuer takes about two-thirds of all fees because it carries credit and fraud risk and funds rewards. Acquirers and processors live on thin basis points and need scale. The 2.6% "drag" is what **A2A rails, QR and BNPL** try to undercut.

Variant B **All-in MDR method: a £200 purchase; total MDR 2.0%; interchange 1.2%; network fee**

0.15%. Find the merchant's net and the acquirer margin.

```
MDR fee          = £200 × 2.0%  = £4.00   → Merchant net = £196.00
Interchange      = £200 × 1.2%  = £2.40   (issuer)
Network fee      = £200 × 0.15% = £0.30   (scheme)
Acquirer margin  = £4.00 − (£2.40 + £0.30) = £1.30  (0.65%)
Issuer share of the fee pool = 2.40 / 4.00 = 60%
```

5 marks **Explain the frictions in correspondent-bank cross-border transfers and how Wise's local-account model lowers cost and time.** CO3

SWIFT correspondent banking has three frictions: ① **multi-hop deductions** (each intermediary bank takes a fee), ② **opaque FX spreads** of 2–4% above the mid-market rate, and ③ **settlement latency** of 3–5 days from batch reconciliation.

Wise holds **local bank accounts in each currency**. The sender pays GBP into Wise's UK account; Wise pays EUR out of its pre-funded Eurozone account. Opposite flows are **netted**, so money doesn't physically cross the border. That removes SWIFT and intermediary fees, allows the **real mid-market rate** with a small transparent fee, and makes transfers near-instant.

60-SECOND RECALL · UNIT 5

- ☐ $1.9T pool; 40% of bank revenue; the Chase usage stats.
- ☐ Name the 5 players and draw the 4-party flow. Three-party example?
- ☐ Recite the $100 waterfall: 1.75 / 0.25 / 0.11 / 0.50 → $97.39; MDR 2.00%; drag 2.61%.
- ☐ Closed-loop vs open-loop vs QR vs A2A.
- ☐ Paytm: 337M/22M, Soundbox, IPO $2.5B @ $18.5B, −27% / −66%, processing = 55–69% of revenue.
- ☐ Wise TAM/SAM/SOM: $20T → $80B → $200M. M-PESA: 2007, Safaricom, SMS/STK, agents.
- ☐ Octopus: 1997, RFID, HKMA 2000, 98%, 15M/day. Stripe 7 lines / $95B vs Adyen $44B.

06 Peer-to-Peer Lending & Crowdfunding

Course outcomes: CO3 (online lending, crowdfunding) · CO5 (business-model analysis)

TIER 2 · CORE

In one line: P2P platforms are **agents, not principals**: they match borrowers and investors for **fees** and push default risk to investors. Retail money proved too small and too flighty, so the winners moved to **institutional money** and then **bought banks**. "Peer-to-peer" became "platform finance".

6.1 Why alternative finance appeared

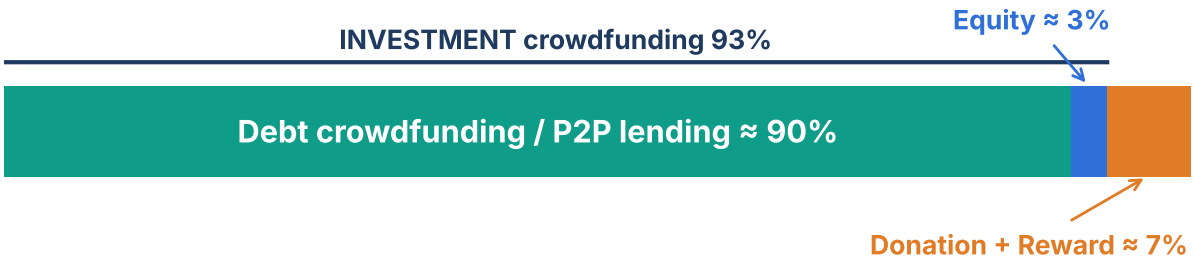
Catalysts

Post-2008 credit freeze (banks pulled back from retail and SME lending under tighter Basel capital rules) · mobile + cloud (AWS 2006) + digital underwriting · collapse of trust in incumbents.

Macro paradigm

Evolution vs revolution again: incumbents saw tech as a delivery pipe; alt-finance platforms aimed to **disintermediate credit gatekeepers** entirely.

6.2 Crowdfunding taxonomy



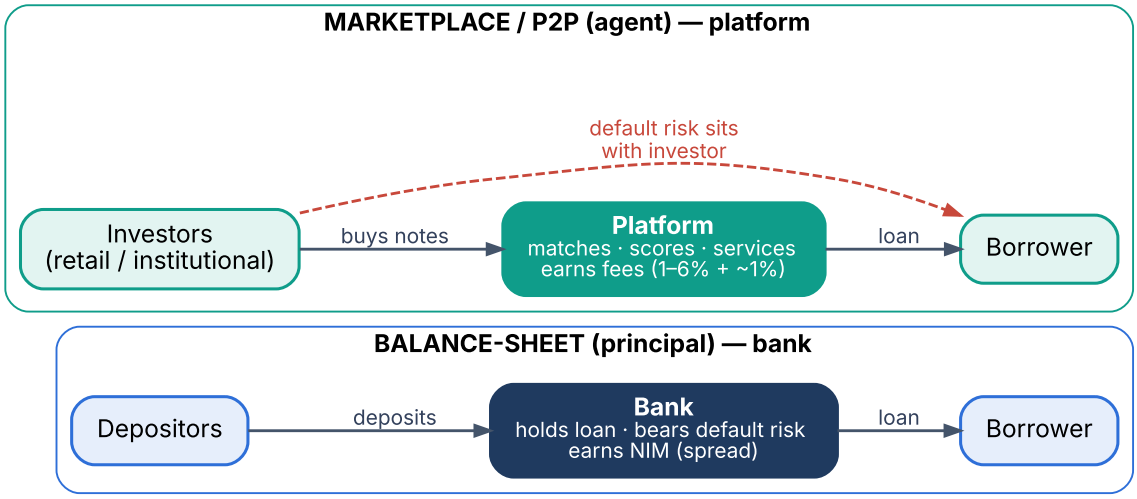
Share of global alternative-finance volume.

Type	Backer gets	Example	Share
INVESTMENT crowdfunding ≈ 93%			
Debt / P2P / marketplace lending	Fractional loan notes paying interest and principal	LendingClub, Funding Circle	≈ 90%
Equity crowdfunding	Shares in a start-up (under regulatory exemptions)	Online equity portals	≈ 3%
NON-INVESTMENT crowdfunding ≈ 7%			
Reward-based	Perks, pre-orders, the product itself	Kickstarter	≈ 7%
Donation-based	Nothing financial (charity, emergencies)	GoFundMe	

MNEMONIC · "D-R-E-D" (BY INCREASING FINANCIAL RETURN)

D onation – nothing back | **R** eward – a product | **E** quity – ownership | **D** ebt – interest

6.3 Economics: balance-sheet bank vs marketplace



Dimension	Balance-sheet lending (bank, principal)	Marketplace / P2P (agent)
Who bears default risk	The bank; losses hit its equity	The investors who bought the notes; the platform holds none (pure model)
Revenue	**Net interest margin** (loan rate − deposit rate)	**Origination fee 1–6%** from borrowers + **servicing fee ~1%** from investors (non-interest income)
Funding	Insured deposits under reserve rules	Retail investors, family offices, hedge and pension funds, securitisation
Maturity transformation	Yes: short deposits → long loans	**None**: capital is locked for the loan term; weak secondary market means an **illiquidity premium**

The five dimensions applied to P2P

- **Information asymmetry:** risky borrowers accept high rates (adverse selection), so platforms use alternative data (cash flow, QuickBooks/Xero, utility bills) and ML to grade loans **A to G** and price each grade.
- **Risk sharing:** unbundled. Each note-holder bears default, but **fractionalisation** lets investors spread across hundreds of notes.
- **Liquidity:** no maturity transformation; limited secondary liquidity.
- **Trust:** moves from bank brands and deposit insurance to **code, scoring transparency and automated servicing**.
- **Transaction costs:** online applications in minutes, money in 24–48 h.

The Lens applied to P2P lending

Who
Subprime-to-prime consumers consolidating debt; SMEs ignored by branch lenders.

What
18%+ credit-card rates; paper applications; weeks of underwriting.

Why
Lower APRs; apply in minutes; cash in 24–48 h.

How
Origination fees at disbursement, investor servicing fees, API licensing.

6.4 Case studies

LendingClub: the pioneer that became a bank

Founded **2006** by **Renaud Laplanche**; launched as a Facebook app for retail P2P. **Note model (pre-charter):**

Borrower applies online → **WebBank** FDIC bank in Utah originates the loan → **LendingClub** buys the loan immediately → **Investors** buy *Member Payment Dependent Notes* mirroring the loan's cash flows

$8.4B+ originations by 2015 on fee income. **2016 crisis:** falsified documents on a **$22M** loan sale to Jefferies → CEO resigned → institutional money (which **multi-homed** across platforms) vanished. **2021:** bought **Radius Bank**, dropped retail P2P notes, took deposits and kept loans on balance sheet.

SoFi: unbundle → rebundle

Stanford, **2011**: alumni funded students' loan refinancing (social affinity, low defaults). Targeted **HENRYs**; used low-margin refinancing as an **acquisition tool** for lifetime value. Rebundled into mortgages, personal loans, robo-advice, crypto and checking. Bought **Golden Pacific Bancorp (2022)** for a charter.

Kickstarter: rewards engine

Founded **2008**. **All-or-nothing:** if the goal isn't hit 100% by the deadline, backers are refunded and no fee is charged. This protects creators from being under-funded and **validates demand**. Fee: **5%** of funds raised + **3–5%** payment processing.

Funding Circle & Lendified

Funding Circle: SME loans funded by institutions and retail. **Lendified:** Canadian SMEs, ML on bank data, decisions in hours vs 4–8 weeks; dual model (balance-sheet lending + **Judi.ai** B2B SaaS). See Unit 3.

SYSTEMIC RISK · THE RISE AND FALL OF CHINESE P2P

2011–2015: served SMEs and consumers excluded by state banks; became the world's largest P2P market with **3,500+ platforms**. **Ezubao:** a Ponzi scheme that raised **$7.6B from 900,000+ investors** with fake borrowers and "guaranteed" **9–14.6%** returns. **Crackdown:** risk reserves, bank custody accounts and a **ban on maturity mismatch**; by **late 2020** virtually all platforms were wound down or converted to micro-lenders.

6.5 Regulation and the end state

United States

Retail P2P notes are **securities** under the Howey Test, so platforms need costly SEC registration (**Form S-1**). **JOBS Act Title III** (Regulation Crowdfunding) opened equity crowdfunding with caps on retail investment.

United Kingdom

The **FCA** created a bespoke regime: credit-assessment rules, **wind-down plans**, marketing limits for high-risk investments.

Retail P2P people fund people → **Institutional marketplace** >80% funded by hedge/pension funds & ABS → **Funding shock** scandal / downturn, capital leaves → **Bank charter** cheap, sticky deposits + NIM + one federal licence

Enduring contribution: algorithmic underwriting, instant API identity checks and frictionless UX are now standard across all credit.

6.6 Exam practice

10 marks **Why did LendingClub and SoFi pivot from retail P2P to institutional funding and then bank charters?** CO3, CO5

1 · Retail P2P constraints

- **High CAC:** retail investors put in $25–$100 per note, expensive relative to capital raised.
- **Supply mismatch:** borrower demand outran retail money.
- **Compliance cost:** notes are securities (SEC S-1 registration).

2 · Institutional phase

Hedge funds, pension funds and securitisation vehicles bought whole loans via API (>80% of originations). Growth followed, but this money is **flighty**: it multi-homes and leaves in downturns or after scandals (LendingClub 2016), and with **no deposits** origination volumes collapse.

3 · Bank charter benefits

1. **Stable, low-cost insured deposits** replace volatile wholesale funding.
2. **Hybrid revenue:** NIM on retained loans + fees on loans sold.
3. **Regulatory efficiency:** one federal charter instead of state-by-state licences.

Examples: LendingClub → Radius Bank (2021); SoFi → Golden Pacific Bancorp (2022). **Conclusion:** the agency model was efficient but fragile; the principal model with FinTech underwriting is more resilient.

5 marks **Contrast balance-sheet and marketplace lending: how each earns revenue and where default risk sits.** CO3, CO5

Balance-sheet (principal): the lender funds loans with its own capital and deposits and holds them. **Revenue** = net interest margin. **Risk** sits on its balance sheet; defaults reduce equity.

Marketplace (agent): the platform matches borrowers with third-party investors. **Revenue** = upfront **origination fees (1–6%)** from borrowers + **servicing fees (~1%)** from investors. **Risk** passes to the note-holders; the pure platform holds none, though it still faces **funding and reputational risk**.

60-SECOND RECALL · UNIT 6

- ☐ 93 / 7 split; 90% debt, 3% equity. D-R-E-D.
- ☐ Agent vs principal: fees 1–6% + ~1% vs NIM. Where does default risk sit?
- ☐ Why P2P has no maturity transformation, and what that means for investors.
- ☐ LendingClub: 2006, Facebook, WebBank notes, $8.4B, 2016 $22M scandal, Radius 2021.
- ☐ SoFi: Stanford 2011, HENRYs, Golden Pacific 2022. Kickstarter: all-or-nothing, 5% + 3–5%.
- ☐ Ezubao: $7.6B, 900k investors, 9–14.6%. US Howey/S-1/JOBS Act vs UK FCA.

07 Robo-Advisors, Open Banking, APIs & BaaS

Course outcomes: CO1 · CO3 (automated wealth) · CO4 (regulation, data privacy)

TIER 1 · HIGH WEIGHT

In one line: Robo-advisors **automate wealth management** (fees 1–2% → 0.25–0.50%), but high CAC means **incumbents and hybrids** won. Open banking makes the **customer the owner of their data**: via **APIs** they can share it with any app, and **BaaS** lets any brand embed a licensed bank inside its own app.

7.1 Wealth management: the value chain

A **$100+ trillion** AUM market with three activities, all being automated:

1 · Financial planning
Goals, risk profiling, tax optimisation, target allocation.

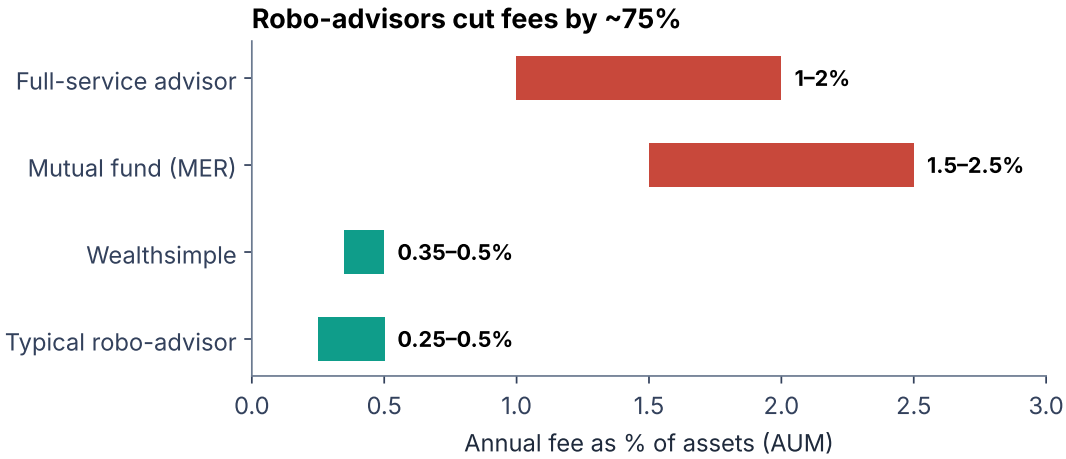
2 · Investing
Portfolio construction, security selection, execution, rebalancing, tax-loss harvesting.

3 · Operations
Onboarding/KYC, custody, reporting, clearing, record-keeping.

7.2 How a robo-advisor works

Questionnaire goals, horizon, risk tolerance → **Risk profile** mapped to a model portfolio → **ETF portfolio** low-cost passive ETFs, mean-variance → **Auto-rebalance** when weights drift → **Tax-loss harvest** sell losers, buy a similar ETF

ETFs are the building block: cheap, liquid, instantly diversified, unlike expensive active mutual funds.



Wealthsimple's seed pitch (a classic Lens example)

Segment	Price	Minimum	Problem
Full-service advisors	~145 bps	$250,000+	Too expensive, excludes young earners
Self-directed brokers	$9–29 / trade	low	No advice; needs effort and knowledge
Wealthsimple (middle market)	**35–50 bps**	**$5,000**	**Automated "light advice" + simple app**

Founded **2014** by **Mike Katchen**, who first built ETF portfolios in Excel for friends; **1,000 clients within months**. Target: educated professionals aged **25–45** in the top income decile without the wealth for private advisors (**HENRYs**), within Canada's **$1.3T** market.

Why pure robos struggled, and who won

The CAC problem
B2C acquisition costs **$500–$1,000 per user**; thin 0.25–0.50% margins need huge AUM → cash burn in the "**valley of death**".

Vanguard fast-follows (2015)
Watched Betterment and Wealthfront, launched its own robo and captured **>50% of robo AUM within two years** using its brand and existing clients. This is Christensen in action: the incumbent responded.

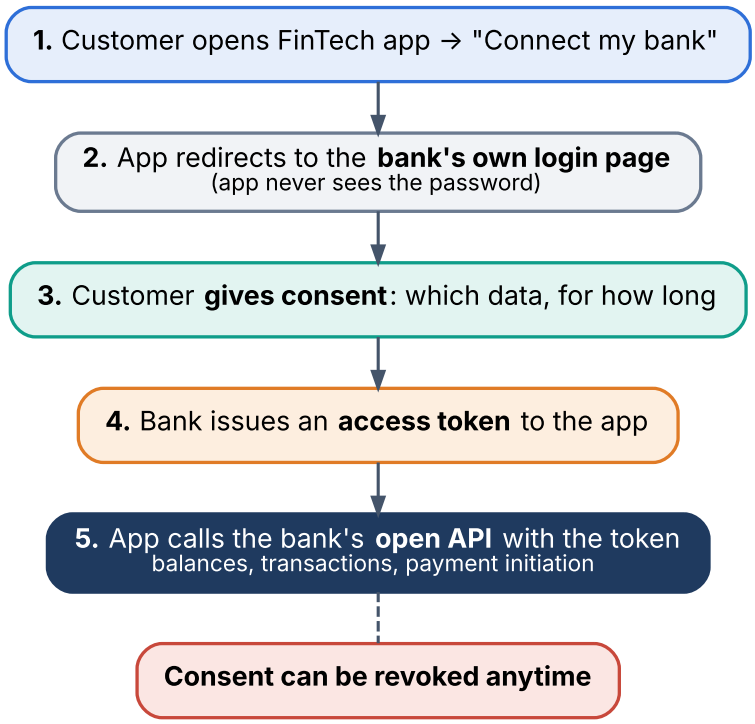
Pivots and hybrids
Start-ups moved to **B2B** (licensing software to banks) or sold out (**Nutmeg → JPMorgan**). **Hybrid "high-tech, high-touch":** algorithms do the routine work; humans handle estate and tax planning and behavioural coaching in drawdowns. B2B WealthTech: **d1g1t**.

7.3 Open banking

Definition: regulation plus technology that lets consumers **control and share** their financial data with third-party apps. The impetus was the post-2008 loss of trust in banks.

EU: PSD2 (2016)
The revised Payment Services Directive forced banks to open account access to registered **third-party providers (TPPs)**: **AISPs** (account information) and **PISPs** (payment initiation). Aim: competition, lower fees, payment innovation.

UK: Open Banking (2016)
Required the largest clearing banks to build **standardised open APIs** for licensed FinTechs with explicit consent. Followed by Australia, Canada and Asia (e.g. India's **Account Aggregator** network).



The consent flow: the app never sees your bank password; it receives a revocable token.

Three types of API

API type	Who can use it	Example use
Closed (internal)	Only inside one institution	Branches ↔ core databases
Partner	Pre-approved partners under contract	Co-branded card with a retailer
Open (public)	Any authorised third party; publicly documented	Budgeting app reading your balances (PSD2)

APIs control **which fields** can be accessed, check credentials and encrypt traffic **without exposing the database**.

Data aggregators (e.g. Plaid) build pipes into thousands of banks so apps can do instant account verification, balance checks, identity checks and transaction categorisation.

7.4 Banking-as-a-Service (BaaS) and embedded finance

A **B2B** model where a **licensed bank exposes** accounts, cards, payments and compliance via cloud APIs, so **non-banks embed banking** in their own app **without a charter**. The bank still owns capital, liquidity, KYC and AML.

LAYER 4 USER	**End customer**: uses banking inside an app they already use (driver, shopper, freelancer).
LAYER 3 DISTRIBUTOR	**FinTech / brand**: owns the UI/UX, customer acquisition and app integration (e.g. a ride-hailing app, a neobank).
LAYER 2 MIDDLEWARE	**BaaS platform**: API translation to the bank's core; ledgers, card issuing, payments routing (e.g. Unit, Treasury Prime).
LAYER 1 LICENCE	**Chartered partner bank**: charter, deposit insurance, central-bank clearing access, AML/KYC compliance.

MNEMONIC · BAAS LAYERS BOTTOM-UP "L-M-D-U"

L icensed bank | **M** iddleware | **D** istributor | **U** ser

"**L**ittle **M**onkeys **D**on't **U**nderstand banking", but they can still offer it.

7.5 Platforms, unbundling and BigTech

- **APIs enable MSPs:** Credit Karma / Borrowell give free scores (subsidy side); banks, lenders and insurers pay referral fees (money side). Same-side effect: more activity → more data. Cross-side: more users → more partners → more choice.
- **Phase 1, unbundling:** start-ups used APIs to peel off single products (micro-investing, FX, budgeting, P2P lending).
- **Phase 2, rebundling:** Revolut and SoFi became one-stop apps; **Nubank** scaled in Latin America on modular cloud with no branches.
- **BigTech / TechFin threat:** Apple, Amazon, Google, Tencent and Ant combine APIs with billions of users. Ant's philosophy: "**Financial services should be like water: turn on the tap and it comes out**",

making bank brands an invisible back-end utility.

- **Outlook:** incumbents mostly **acquired or partnered** rather than died; the industry is converging on API-driven open ecosystems.

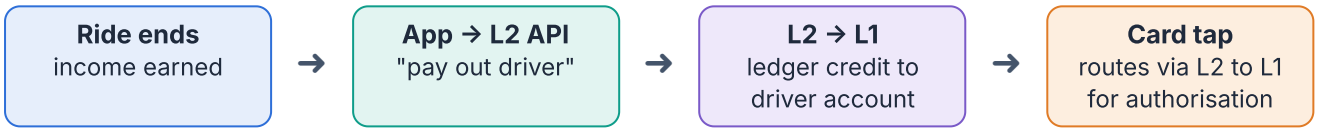
7.6 Exam practice

10 marks **A ride-hailing app wants to give drivers bank accounts and debit cards in-app. Design it with the 4-layer BaaS model and assess compliance risks.** CO1, CO3, CO4

1 · Layer-by-layer

- **L1 · Partner bank:** charter, deposit insurance, clearing access; holds driver deposits; runs KYC, AML and sanctions screening.
- **L2 · BaaS middleware** (e.g. Unit, Treasury Prime): REST APIs to the bank core; ledger, virtual and physical card issuing, payment routing.
- **L3 · Ride-hailing company:** builds the wallet screens (balance, earnings, instant payout) into its own app.
- **L4 · Drivers:** get paid instantly and spend with a branded debit card, with no branch visit.

2 · Money flow



3 · Risks

- **Custody / commingling:** keep company funds and driver deposits strictly segregated.
- **Disclosure:** drivers must know the **bank**, not the app, provides the account and insurance.
- **Third-party / concentration risk:** the brand relies on the middleware and bank; the bank remains liable for compliance failures made through partners.

Value: new revenue for the platform (interchange share on debit spend), driver loyalty, and inclusion for under-banked gig workers.

10 marks **Compare the EU's PSD2 with consent-based Account Aggregator models (e.g. India) on API standards, privacy, hurdles and innovation.** CO1, CO3, CO4

Dimension	EU PSD2	Account Aggregator (AA) model
Approach	Statutory mandate: banks must open APIs to licensed TPPs	Central-bank-regulated **consent architecture** with licensed data intermediaries
Data flow	Direct: bank ↔ TPP	Indirect: bank → AA (consent manager) → FinTech
API standards	Varied across banks and countries at first → integration friction	Common technical standards → one integration reaches many institutions
Consent & privacy	Given to each app at linking; governed by GDPR	Central, granular, revocable consent; AA is **data-blind** (routes encrypted data, doesn't store it)

Evaluation: PSD2's mandate removed incumbent resistance quickly, but early non-standard APIs raised costs for start-ups. The AA model's standardisation and consent layer lower integration cost

and strengthen user control, which speeds up aggregation apps, cash-flow underwriting and personal finance tools. Its weakness is that it depends on ecosystem adoption by data providers.

60-SECOND RECALL · UNIT 7

- ☐ Three wealth activities; the five-step robo workflow; why ETFs.
- ☐ Wealthsimple: 145 bps & $250k vs 35–50 bps & $5k; Katchen 2014; HENRYs 25–45.
- ☐ CAC $500–1,000; Vanguard >50% robo AUM in 2 years; Nutmeg → JPMorgan; hybrid; d1g1t.
- ☐ PSD2 and UK Open Banking (2016); AISP vs PISP; draw the consent flow.
- ☐ Closed vs partner vs open APIs; what Plaid does.
- ☐ BaaS L-M-D-U with an example at each layer; Ant's "water" quote.

Night-before cheat sheet

Everything to have in your head walking into the exam

All mnemonics on one line each

Mnemonic	Stands for	Unit
C-P-V-M	Customer · Pain point · Value proposition · Monetisation (the Lens: "Can People Value Money?")	1
C-S-G-M	Cloud (2006) · Smartphones (2007–08) · GFC · Millennials: the FinTech 3.0 catalysts	1
I-T-L-R-T	Information asymmetry · Transaction costs · Liquidity creation · Risk sharing · Trust	2
R-R-D-A	Pisano: Routine (old/old) · Radical (new tech) · Disruptive (new model) · Architectural (both new)	2
S-U-R	Supervised = scoring · Unsupervised = unusual (AML) · Reinforcement = rewards (trading)	3
F-DOAC	SHA-256: Fixed 256-bit · Deterministic · One-way · Avalanche · Collision-resistant	4
M-A-I-C	Card players: Merchant · Acquirer · Issuer · Cardholder (+ Network in the middle)	5
D-R-E-D	Crowdfunding: Donation · Reward · Equity · Debt	6
L-M-D-U	BaaS, bottom-up: Licensed bank · Middleware · Distributor · User	7

Formulas you might have to use

```
CARD FEES   MDR = interchange + acquirer surcharge          ($100: 1.75 + 0.25 = $2.00)
            Total drag = MDR + network + processing           (2.00 + 0.11 + 0.50 = $2.61)
            Merchant net = gross − total fees                  ($97.39)
            All-in MDR variant: acquirer margin = MDR − interchange − network fee
PLATFORM    Revenue = GMV × take rate        (pure processing ≈ 10–20 bps)
MARKET SIZE TAM → SAM → SOM   Wise: $20T → $2T×4% = $80B → 50%×$80B×0.5% = $200M
BITCOIN     PoW: SHA-256(header+nonce) < T        D = T_max / T
            T_new = T_old × (actual minutes for 2,016 blocks / 20,160)
            TPS ≈ txns per block / 600 s ≈ 1,500/600 ≈ 2.5
            Supply: 50 → 25 → 12.5 → 6.25 … every 210,000 blocks → 21M cap
FORK        Theory: P_pre = P_post + P_spin-off
            Reality (BCH): 2,741 + 380 = 3,121 vs 2,746 → +13.6%
RISK        Sharpe = μ / σ    crypto 20.44/70.80 = 0.29   vs   US stocks 0.94/3.42 = 0.27
FX SAVING   Legacy $2,625 vs FinTech $375 on $100k → saves $2,250 (85.7%)
```

Numbers worth quoting

Fact	Number
FinTech investment 2008–22	$1.2T / ~38,000 deals
Wave peaks: 2016 / 2019 / 2021	$73.7B / >$200B / ~$240B

Fact	Number
Bitcoin TPS vs Visa	2–4 vs 24,000
Bitcoin energy; PoS saving	~73 TWh (Austria); −99.95%
ICOs 2014–Jul 2018	860 / $20.07B

Unbanked adults; gained accounts	1.7B; 1.2B (−35%)
Bank intermediation cost since 1880s	~2% (sticky)
JPMorgan principal / agency	65% / 35%
FinTechs agency share	90–100%
Payments revenue pool	$1.9T (≈40% of bank revenue)
Insurance value chain / wealth AUM	$6T / $100T
Start-up CAC per user	$500–$1,000
Start-up failure rate	~90%

BTC co-movement explains altcoin variance	~80%
Alt-finance: debt / equity / non-invest.	90% / 3% / 7%
P2P fees: origination / servicing	1–6% / ~1%
Institutional share of marketplace loans	>80%
Advisor vs robo fees	1–2% vs 0.25–0.50%
Vanguard robo share in 2 yrs	>50%
Lemonade: Maya / Jim	2 min / 3 sec

Case → concept map (which example proves which idea)

Case	Use it to illustrate
Wise	The Lens; disintermediation of SWIFT; netting; TAM/SAM/SOM; Group 2 social-good FinTech
Credit Karma / Borrowell	Multi-sided platform: subsidy vs money side; B2B2C
Paytm / Alipay / Ant	Zero-fee QR subsidy side → cross-sell credit and wealth; TechFin; thin take-rate economics; IPO de-rating
M-PESA	Financial inclusion; mobile rails without banks; agent networks
Octopus	Closed-loop → general stored value; regulatory licence as an enabler
Stripe / Adyen	Acquiring as software (APIs); valuation multiples
Lendified / Judi.ai	ML underwriting with alternative data; B2B SaaS pivot
Lemonade / ZhongAn	AI in InsurTech; cloud-native micro-policies; principal-based exception (ZhongAn 96%)
Bitcoin / BCH fork	Double-spend solution; PoW; irrational pricing
Ripple / FTX	Alternative consensus and regulatory (Howey) risk / centralised governance failure and moral hazard
LendingClub / SoFi	Agent → principal pivot; funding fragility; bank charters; unbundle → rebundle
Kickstarter / Ezubao	All-or-nothing reward design / P2P fraud and systemic risk
Wealthsimple / Vanguard	Lens for robo-advice / incumbent fast-follower beating start-ups (Christensen)
Plaid / Nubank	API aggregation layer / branchless rebundled neobank

Acronyms

PSD2 Payment Services Directive 2 · **TPP / AISP / PISP** third-party provider / account-information / payment-initiation service provider · **API** application programming interface · **BaaS / SaaS / PaaS / IaaS** banking / software / platform / infrastructure-as-a-service · **P2P** peer-to-peer · **MSP** multi-sided platform · **NIM** net interest margin · **MDR** merchant discount rate · **A2A** account-to-account · **GMV** gross merchant volume · **BNPL** buy now pay later · **SWIFT** interbank messaging network · **ACH** automated clearing house (batch net settlement) · **KYC / AML** know your customer / anti-money laundering · **XAI, SHAP, LIME** explainable AI tools · **NLP** natural language processing · **CAC** customer acquisition cost · **HENRY** high earner, not rich yet · **AUM / AUA** assets under management / administration · **MER** management expense ratio · **ETF** exchange-traded fund · **DLT** distributed ledger technology · **PoW / PoS** proof-of-work / proof-of-stake · **SHA-256** secure hash algorithm 256-bit · **TPS** transactions per second · **ICO** initial coin offering · **DeFi, dApp, DEX, AMM, TVL, NFT** (see Unit 4) · **CBDC** central bank digital currency · **TAM / SAM / SOM** total / serviceable available / serviceable obtainable market · **EV/Sales, EV/EBITDA, P/E** valuation multiples · **VC** venture capital · **FFF** friends, family & fools.

Final readiness checklist

☐ **Paradigms:** contrast evolutionary vs revolutionary with business-model examples.
☐ **Lens:** apply C-P-V-M to any new venture in under two minutes.
☐ **Intermediation:** explain all five dimensions, plus adverse selection vs moral hazard.
☐ **Strategy:** draw Christensen and Pisano; place real FinTechs on them.
☐ **Platforms:** subsidy vs money side; same-side vs cross-side network effects; agency vs principal.
☐ **ML:** pick the right paradigm for a use case; explain the XAI trade-off and fair-lending rules.
☐ **Blockchain:** hash properties, chain linking, PoW condition, difficulty, halving, PoW vs PoS, forks.
☐ **Payments:** do the $100 waterfall and the all-in MDR variant without notes.
☐ **Lending:** P2P economics, why platforms bought banks, D-R-E-D, Chinese P2P collapse.
☐ **Open banking & BaaS:** PSD2, three API types, consent flow, L-M-D-U, robo-advisor economics.

> **If you blank in the exam:** fall back on the Lens (who, what, why, how), then one intermediation dimension, then a trade-off. That structure earns marks on almost any FinTech question.